

Penalized and Debiased GMM with Potentially Invalid Moment Conditions

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Abstract

This paper proposes a penalized debiased generalized method of moments (PDG) procedure for estimation and moment selection when candidate moment conditions may be invalid and the moment functions depend on first-step nonparametric or machine-learning estimators. We assume that a set of valid baseline moments identifies the parameter of interest, while additional candidate moments may improve estimation precision when valid. To account for the plug-in step bias, we construct orthogonal and cross-fitted sample debiased moments for potentially invalid moments, building on [Chernozhukov et al. \(2022a\)](#). We show that the proposed procedure has oracle-like properties: it consistently selects the correct moment conditions, and its asymptotic distribution is the same as that of the infeasible oracle GMM, even when the moments contain estimated first-step components, for a fixed data-generating process. The simulation results show that PDG accurately recovers the invalid set while exhibiting finite-sample shrinkage bias, and that the post-selection PDG, which we recommend for final estimation and inference, has negligible bias and provides substantial variance reductions relative to baseline-only GMM.

1 Introduction

This paper studies moment selection in semiparametric models in which the moment functions depend on a finite-dimensional parameter of interest and unknown nuisance functions estimated in a first step. Additional moment conditions can improve the precision of generalized method of moments (GMM) estimation, but their validity is not always known in advance. We consider a setting in which a set of valid baseline moments identifies the parameter of interest, while additional candidate moments may or may not be valid. Using only the baseline moments avoids imposing invalid candidate restrictions, but may discard useful information. In contrast, treating all candidate moments as valid can lead to an inconsistent estimator. This problem arises, for example,

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in instrumental-variable models with a set of valid instruments and additional instruments whose exogeneity is uncertain. The objective is to exploit the information in the valid candidate moments without knowing in advance which candidates are valid, while accounting for first-step estimation of the nuisance functions.

First-step estimation introduces an additional challenge for moment selection. In the partially linear instrumental-variable model considered in this paper, for example, constructing the moment functions involves estimating conditional expectations. These functions can be estimated using non-parametric or machine-learning methods rather than imposing a parametric specification. However, first-step estimation errors enter the sample moments and can affect both moment selection and the asymptotic distribution of the estimator. Consistency of the first-step estimators alone does not ensure that these effects are negligible. A moment-selection procedure in this setting must therefore account for first-step estimation while distinguishing valid and invalid candidate moments.

To address this problem, we propose a penalized debiased GMM (PDG) procedure. Following [Liao \(2013\)](#), we introduce an auxiliary parameter for each candidate moment to represent its population mean at the true parameter. A zero auxiliary parameter corresponds to a valid moment, whereas a nonzero auxiliary parameter corresponds to an invalid moment. This gives an augmented moment system that is correctly specified without requiring every candidate restriction to be valid. Building on [Chernozhukov et al. \(2022a\)](#), we construct orthogonal augmented moments by adding first-step influence-function adjustments and use cross-fitting to form their sample counterparts. We then jointly estimate the parameter of interest and the auxiliary parameters by minimizing a GMM criterion with a penalty on the auxiliary parameters.

The main contribution of this paper is to establish that consistent moment selection and the oracle debiased GMM asymptotic distribution are preserved in the presence of nonparametric or machine-learning first-step estimators. Under the stated regularity and rate conditions, orthogonalization and cross-fitting make the effect of replacing the nuisance functions by estimates negligible in the debiased sample moment expansion at the $\sqrt{n}$ scale. We establish consistency, the convergence rate, and oracle properties of the PDG estimator under appropriate conditions on the penalty. In particular, the procedure identifies the valid candidate set with probability approaching one and, under optimal weighting, has the same first-order asymptotic distribution as the infeasible debiased GMM estimator that knows the valid moment set in advance. The analysis maintains a fixed number of moments, and the oracle results are pointwise for a fixed data-generating process.

Simulations in a partially linear instrumental-variable model show high moment-selection accuracy at $n = 1000$ and $n = 2000$, with the unpenalized post-selection refit (Post-PDG) exhibiting small bias and root mean squared error close to the oracle benchmark. Its empirical variance reductions relative to baseline-only GMM are approximately 82% in designs with lower baseline-instrument relevance and 26–28% in designs with higher relevance, closely matching the population oracle comparisons. Post-PDG also substantially reduces the finite-sample bias observed in direct PDG, supporting its use for final estimation and inference.

1.1 Literature Review

This paper is related to the literature on GMM moment selection (Andrews, 1999; Andrews and Lu, 2001; Hong et al., 2003). Our method builds on the penalized moment-selection framework of Liao (2013) and is related to subsequent developments in Cheng and Liao (2015); Caner et al. (2018); Chang et al. (2023). We retain the distinction between valid baseline moments that identify the parameter of interest and potentially invalid candidate moments, but focus on moment functions containing flexibly estimated first-step components.

The paper also relates to semiparametric two-step GMM (Newey, 1994; Akerberg et al., 2014) and the literature on orthogonal moments and debiased machine learning (Chernozhukov et al., 2018, 2022a). Building on Chernozhukov et al. (2022a), we develop an orthogonalization construction for the augmented system with potentially invalid candidate moments and combine it with cross-fitting and penalization. Our contribution is to establish that consistent moment selection and the pointwise oracle debiased GMM asymptotic distribution are preserved in the presence of nonparametric or machine-learning first-step estimators, under the stated conditions.

A closely related paper is Ouyang and Yang (2025), who extend penalized moment selection to a semiparametric binary-choice model with a nonparametric first-step plug-in. While their analysis is developed for a specific binary-choice model, we provide a general framework for moment selection in semiparametric GMM. Our framework is not restricted to a particular model and accommodates flexible nonparametric or machine-learning first-step estimators under the stated conditions.

The remainder of the paper is organized as follows. Section 2 introduces the model, constructs the PDG estimator, and provides two illustrative examples. Section 3 establishes the asymptotic properties of the estimator and discusses implementation, inference, and post-selection estimation. Section 4 presents the simulation results, and Section 5 concludes.

2 Penalized Debiased GMM

This section introduces the model and formally defines the penalized debiased generalized method of moments (PDG) estimator. Throughout, unless otherwise stated, derivatives with respect to a scalar variable t are evaluated at $t = 0$, expectations without subscripts are taken under the true distribution F_0 , all vectors are column vectors, and, for any two vectors x_1 and x_2 , (x_1, x_2) denotes $(x_1^T, x_2^T)^T$.

2.1 Model and Augmented Moments

Let θ denote a d_θ -dimensional parameter of interest, let γ denote an unknown first-step function, and let W denote a random vector of observed variables with distribution F_0 . Suppose that an m_1 -dimensional moment function $g_1(W, \gamma, \theta)$ satisfies

$$E[g_1(W, \gamma_0, \theta_0)] = 0, \tag{1}$$

where γ_0 is the probability limit under F_0 of a first-step estimator $\hat{\gamma}$. We assume that these moment conditions identify θ_0 ; that is, θ_0 is the unique solution to $E[g_1(W, \gamma_0, \theta)] = 0$. Under suitable regularity conditions, θ_0 can be estimated by standard semiparametric two-step methods; see, for example, [Newey and McFadden \(1994, Section 8\)](#) and [Newey \(1994\)](#).

Suppose further that there exists an m_2 -dimensional moment function $g_2(W, \gamma, \theta)$ such that

$$E[g_2(W, \gamma_0, \theta_0)] = \mu_0, \quad (2)$$

where $\mu_0 = (\mu_{0,1}, \dots, \mu_{0,m_2})$ is unknown, and each component $\mu_{0,j}$ may or may not be equal to zero. When $\mu_{0,j} = 0$, the moment condition $E[g_{2,j}(W, \gamma_0, \theta_0)] = 0$ is valid. Under optimal GMM weighting, augmenting the baseline moments with a valid additional moment weakly reduces the asymptotic covariance matrix of the estimator of θ_0 , with no efficiency gain if and only if the additional moment is redundant for estimating θ_0 ; see [Hansen \(1982\)](#) and [Breusch et al. \(1999\)](#). By contrast, when $\mu_{0,j} \neq 0$, imposing the restriction $E[g_{2,j}(W, \gamma_0, \theta_0)] = 0$ introduces model misspecification and may render the estimator of θ_0 inconsistent.

The objective is to estimate θ_0 efficiently by exploiting all valid moment conditions in g_2 while accounting for first-step estimation of γ_0 . This requires consistently selecting the valid moments in g_2 , which amounts to recovering the zero and nonzero components of μ_0 . Following [Liao \(2013\)](#), we introduce an auxiliary parameter vector $\mu \in \mathbb{R}^{m_2}$ and define the augmented moment function

$$g^*(W, \gamma, \theta, \mu) := \begin{bmatrix} g_1(W, \gamma, \theta) \\ g_2(W, \gamma, \theta) - \mu \end{bmatrix}. \quad (3)$$

At the true value, the zero components of μ_0 correspond to valid additional moments. The parameter vector is therefore augmented from θ to (θ, μ) . At (θ_0, μ_0) ,

$$E[g^*(W, \gamma_0, \theta_0, \mu_0)] = 0. \quad (4)$$

2.2 Orthogonal Moment Functions

When the first-step function γ_0 is known, the methods of [Liao \(2013\)](#) and [Cheng and Liao \(2015\)](#) jointly estimate θ_0 and select the valid moments in g_2 using penalized GMM. Under suitable regularity conditions and optimal weighting, these procedures attain the same first-order asymptotic distribution as the infeasible oracle estimator that uses all and only the valid moments. In the present setting, however, γ_0 is unknown and must be replaced by a nonparametric or machine-learning estimator $\hat{\gamma}$. Consequently, the effect of first-step estimation on the asymptotic properties of the resulting estimator must be taken into account, as in [Newey \(1994\)](#). To address this issue, the method proposed in this paper builds on recent work on debiased and orthogonal moment conditions ([Chernozhukov et al., 2018, 2022a,c,d](#)) and accommodates a generic first-step estimator $\hat{\gamma}$ that may be complex or high dimensional.

In particular, we construct a debiased version of the augmented moment function $g^*(W, \gamma, \theta, \mu)$

that accommodates potentially invalid moment conditions and a flexibly estimated first step. Let F denote a generic distribution of W , and let $\gamma(F)$ denote the probability limit of $\hat{\gamma}$ when W has distribution F , allowing for possible misspecification. We leave F unrestricted apart from regularity conditions ensuring that $\gamma(F)$ is well defined. In particular, $\gamma_0 = \gamma(F_0)$.

For an alternative distribution H , let $F_t = (1 - t)F_0 + tH$, $t \in [0, 1]$, denote a path from F_0 to H . We restrict attention to distributions H such that $\gamma(F_t)$ exists for all sufficiently small t and the remaining regularity conditions are satisfied. Following Chernozhukov et al. (2022a), we assume that, for each $q = 1, 2$, there exists a function $\phi_q(w, \gamma, \alpha, \theta)$ such that, for every admissible H and every $\theta \in \Theta$,

$$\frac{d}{dt}E[g_q(W, \gamma(F_t), \theta)] = \int \phi_q(w, \gamma_0, \alpha_0, \theta) H(dw), \quad (5)$$

and

$$E[\phi_q(W, \gamma_0, \alpha_0, \theta)] = 0, \quad (6)$$

with $E[\|\phi_q(W, \gamma_0, \alpha_0, \theta)\|^2] < \infty$. The derivative is taken from the right at $t = 0$. Here ϕ_q is the first-step influence function (FSIF), α is an additional nuisance function on which ϕ_q depends, and α_0 denotes its true value (Ichimura and Newey, 2022; Chernozhukov et al., 2022a).

The preceding condition provides an influence-function representation of the Gateaux derivative of the map $F \mapsto E[g_q(W, \gamma(F), \theta)]$. Under suitable regularity conditions, the existence of a regular estimator of this functional at the $\sqrt{n}$ rate implies pathwise differentiability and hence the existence of an FSIF (van der Vaart, 1991).

Note that the FSIF $\phi_2(W, \gamma, \alpha, \theta)$ does not depend on the auxiliary parameter μ . Indeed, because μ is held fixed along the path F_t ,

$$\frac{d}{dt}E[g_2(W, \gamma(F_t), \theta) - \mu] = \frac{d}{dt}E[g_2(W, \gamma(F_t), \theta)]. \quad (7)$$

Thus, introducing the auxiliary parameter μ to accommodate potentially invalid moment conditions does not alter the construction of the FSIF for the augmented moment function. We refer the reader to Chernozhukov et al. (2022a), Chernozhukov et al. (2022c), and Chernozhukov et al. (2022d) for further discussion of how to derive the FSIF for general moment functions and estimate the additional nuisance function α .

Let $\phi(W, \gamma, \alpha, \theta) = (\phi_1(W, \gamma, \alpha, \theta), \phi_2(W, \gamma, \alpha, \theta))$. The debiased augmented moment function is defined as

$$\psi^*(W, \gamma, \alpha, \theta, \mu) := \begin{bmatrix} g_1(W, \gamma, \theta) \\ g_2(W, \gamma, \theta) - \mu \end{bmatrix} + \begin{bmatrix} \phi_1(W, \gamma, \alpha, \theta) \\ \phi_2(W, \gamma, \alpha, \theta) \end{bmatrix} = g^*(W, \gamma, \theta, \mu) + \phi(W, \gamma, \alpha, \theta). \quad (8)$$

Let $g = (g_1, g_2)$ and $\psi = g + \phi$. When $\mu = 0$, ψ^* reduces to the debiased moment function ψ proposed in Chernozhukov et al. (2022a). By construction,

$$E[\psi^*(W, \gamma_0, \alpha_0, \theta_0, \mu_0)] = E[g^*(W, \gamma_0, \theta_0, \mu_0) + \phi(W, \gamma_0, \alpha_0, \theta_0)] = 0. \quad (9)$$

Hence, ψ^* can be used to construct estimators of θ_0 and μ_0 from an i.i.d. sample $W_1, \dots, W_n$, where $W_i \sim F_0$.

Intuitively, the construction of the debiased augmented moment function ψ^* may be viewed as analogous to residualizing the original moment function g with respect to the nuisance tangent space generated by perturbations of γ . In this sense, ψ^* is orthogonal to the nuisance directions associated with the first-step function. More formally, for any admissible direction δ along which γ departs locally from γ_0 ,

$$\frac{d}{dt} E[\psi^*(W, \gamma_0 + t\delta, \alpha_0, \theta, \mu)] = 0 \quad (10)$$

for all $\theta \in \Theta$. This identity shows that adding the FSIF makes the debiased augmented moment function locally insensitive to the first-order effect of estimation error in γ . Because introducing the auxiliary parameter μ does not alter the form of the FSIF, this orthogonality condition is essentially the same as that established in Theorem 3 of Chernozhukov et al. (2022a). In addition, we show in the theory section that this orthogonality condition does more than provide local robustness to the plug-in first step: it also allows the estimator to retain the same oracle properties as in the parametric case, even when γ_0 is replaced by a flexibly estimated first-step function.

The construction of the debiased augmented moment function requires estimating an additional nuisance function α , whose form depends on the original moment function g . Under the conditions of Theorem 2 in Chernozhukov et al. (2022a), the FSIF is globally robust to variation in α when $\gamma = \gamma_0$, in the sense that

$$E[\phi(W, \gamma_0, \alpha, \theta)] = 0 \quad (11)$$

for all $\theta \in \Theta$ and all α in an admissible set $\mathcal{A}$ containing α_0 . Consequently, the population debiased augmented moment is invariant to α at γ_0 :

$$E[\psi^*(W, \gamma_0, \alpha, \theta, \mu)] = E[g^*(W, \gamma_0, \theta, \mu)]. \quad (12)$$

2.3 Cross-Fitting and Penalized Estimation

In addition to constructing orthogonal moments, we employ cross-fitting to control the asymptotic remainder terms arising from first-step estimation, as in Newey and Robins (2018), Chernozhukov et al. (2018), and Chernozhukov et al. (2022a). Cross-fitting is now widely used for estimators involving nonparametric or machine-learning first steps because it avoids strong empirical-process restrictions, such as Donsker conditions, on the underlying function classes and simplifies the regularity conditions required for asymptotic analysis. Combined with orthogonalization and suitable convergence-rate conditions on the first-step estimators, cross-fitting ensures that the effect of first-step estimation on the sample debiased augmented moments is asymptotically negligible at the $\sqrt{n}$ scale. As a result, the feasible moments have the same first-order behavior as the corresponding oracle moments evaluated at the true nuisance functions. This property is central to our analysis because it allows the shrinkage estimator to retain the oracle properties obtained in the parametric setting even when γ_0 is replaced by a flexibly estimated first-step function.

In particular, let $W_1, \dots, W_n$ be an i.i.d. sample, and partition the index set $\{1, \dots, n\}$ into L disjoint folds $I_1, \dots, I_L$. For each fold $\ell = 1, \dots, L$, let $\hat{\gamma}_\ell$, $\hat{\alpha}_\ell$, and an initial estimator $\tilde{\theta}_\ell$ be constructed using only observations with indices outside I_ℓ . The cross-fitted debiased augmented sample moment function is defined as

$$\bar{\psi}^*(\theta, \mu) := \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \left[g^*(W_i, \hat{\gamma}_\ell, \theta, \mu) + \phi(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell) \right]. \quad (13)$$

The initial estimator $\tilde{\theta}_\ell$ can be constructed from the baseline moment function g_1 using only observations outside I_ℓ and may also be used below in constructing certain data-dependent penalty functions for the auxiliary parameters. The choice of the first-step estimator $\hat{\gamma}_\ell$ is application specific and may be based on standard nonparametric or machine-learning methods. When the form of α_0 is known, $\hat{\alpha}_\ell$ can be constructed by estimating the nuisance components entering its analytical expression. When such an expression is unavailable or difficult to derive, α_0 may instead be estimated using automatic methods; see [Chernozhukov et al. \(2022c\)](#). In the asymptotic analysis below, we impose high-level consistency and convergence-rate conditions on these preliminary estimators.

When some moment conditions in g_2 are potentially invalid, the corresponding components of μ_0 are nonzero, but their locations are unknown. Imposing all moment conditions in (g_1, g_2) as valid and minimizing the resulting GMM criterion with respect to θ generally yields an inconsistent estimator, even after the debiasing adjustment. Following [Liao \(2013\)](#) and [Cheng and Liao \(2015\)](#), we recast the joint estimation and moment selection problem as a variable selection problem by incorporating a penalty on the components of μ into the GMM criterion. Specifically, let $\hat{P}_{\lambda_n}(\mu_j)$ denote a generic, possibly data-dependent penalty applied to the j th component of μ , where the tuning parameter λ_n controls the strength of the penalty. For the remainder of this paper, we suppress the subscript λ_n to simplify notation, with the dependence on λ_n made explicit in the examples below. The penalized debiased generalized method of moments (PDG) estimator is defined as

$$(\hat{\theta}, \hat{\mu}) := \arg \min_{\theta, \mu} \bar{\psi}^*(\theta, \mu)^T V_n^* \bar{\psi}^*(\theta, \mu) + \sum_{j=1}^{m_2} \hat{P}(\mu_j), \quad (14)$$

where V_n^* is an $(m_1 + m_2) \times (m_1 + m_2)$ symmetric positive definite weighting matrix.

The penalty $\hat{P}(\mu_j)$ is designed to distinguish the zero and nonzero components of μ_0 . It should impose strong shrinkage on components close to zero while exerting relatively little influence on components bounded away from zero. With an appropriate choice of the tuning parameter, this separation can produce the oracle property studied in the variable selection literature ([Fan and Li, 2001](#)). The oracle property consists of two parts: selection consistency, whereby the zero and nonzero components are correctly distinguished with probability approaching one, and an oracle limiting distribution, whereby the estimator of the nonzero components has the same first-order distribution as the infeasible estimator that knows the true support in advance. Standard penalties with this property include the bridge penalty ([Knight and Fu, 2000](#)), the adaptive Lasso ([Zou,](#)

2006), and the SCAD penalty (Fan and Li, 2001).

A useful geometric intuition is that a penalty with the oracle property should have a sharp corner or singularity at the origin, thereby encouraging small coefficients to be estimated as exactly zero, while becoming relatively flat away from the origin to reduce the shrinkage bias on nonzero coefficients. The bridge and SCAD penalties exhibit this geometry through their concavity as functions of $|\mu_j|$. The adaptive Lasso achieves a similar separation through data-dependent penalty slopes: the weight is large when the corresponding initial estimate is close to zero and remains bounded when its probability limit is nonzero.

The next section states general conditions on $\hat{P}$ and rate conditions for the tuning parameter λ_n that establish the convergence rate and oracle properties of the PDG estimator. After developing the asymptotic theory, we discuss implementation under the adaptive Lasso penalty (Zou, 2006), defined by

$$\hat{P}(\mu_j) = \frac{\lambda_n}{|\tilde{\mu}_j|^r} |\mu_j|, \quad (15)$$

where $r > 0$ and $\tilde{\mu}_j$ is an initial estimator of $\mu_{0,j}$. We then provide a step-by-step implementation procedure. For other penalty functions, we refer readers to Appendix B of Liao (2013).

Before proceeding to the asymptotic theory, we provide two examples illustrating how debiased augmented moment conditions can be constructed in concrete settings.

Example 1 (Endogenous Partially Linear Model). Consider the partially linear regression model studied by Robinson (1988),

$$Y = X\theta_0 + h_0(U) + \epsilon, \quad (16)$$

where θ_0 is the true value of the parameter of interest θ , Y and X are scalar random variables, U is a random vector that affects Y through an unknown function h_0 , and ϵ is the stochastic error. Suppose that

$$E[\epsilon | U] = 0. \quad (17)$$

The model implies that

$$Y - E[Y | U] = \{X - E[X | U]\}\theta_0 + \epsilon. \quad (18)$$

Suppose that a valid baseline instrument Z_1 and a candidate instrument Z_2 are available. For identification, suppose that

$$E[\{Z_1 - E[Z_1 | U]\}\{X - E[X | U]\}] \neq 0. \quad (19)$$

In addition, suppose that

$$E[Z_1\epsilon] = 0 \quad \text{and} \quad E[Z_2\epsilon] = \mu_0, \quad (20)$$

where μ_0 is the true value of an auxiliary parameter μ that measures the violation of the candidate moment condition. Following the calculations in Newey (1994) or Ichimura and Newey (2022), the

augmented debiased moment conditions are

$$\begin{aligned} E[\{Z_1 - E[Z_1 | U]\}(\{Y - E[Y | U]\} - \{X - E[X | U]\}\theta)] &= 0, \\ E[\{Z_2 - E[Z_2 | U]\}(\{Y - E[Y | U]\} - \{X - E[X | U]\}\theta)] - \mu &= 0. \end{aligned} \quad (21)$$

Example 2 (Endogenous Single-Index Model). Consider the endogenous single-index model studied by [Birke et al. \(2017\)](#),

$$Y = h_0(X^T \theta_0) + \epsilon, \quad E[\epsilon | Z_1] = 0, \quad (22)$$

where θ_0 is the true value of the parameter of interest θ , Y is a scalar random variable, X is a vector of potentially endogenous regressors, Z_1 is a vector of valid instruments, h_0 is an unknown function, and ϵ is the stochastic error. Let Θ impose the usual scale and sign normalization on θ . For each $\theta \in \Theta$, suppose that there exists a unique profile NPIV solution h_θ satisfying

$$E[Y - h_\theta(X^T \theta) | Z_1] = 0, \quad h_{\theta_0} = h_0. \quad (23)$$

Suppose that a vector of valid baseline instruments Z_2 and a vector of candidate instruments Z_3 are available and satisfy

$$E[Z_2 \epsilon] = 0 \quad \text{and} \quad E[Z_3 \epsilon] = \mu_0, \quad (24)$$

where μ_0 is the true value of an auxiliary parameter μ that measures violations of the candidate moment conditions. The maintained-valid instrument vector Z_1 is used to estimate h_θ , while Z_2 supplies additional valid restrictions for identifying θ_0 , since the restrictions generated by Z_1 are already imposed in the profile NPIV first step; Z_3 is excluded from the first step because its validity is unknown. Assume that the baseline moments locally identify θ_0 ; see [Chen et al. \(2014\)](#) for related rank and completeness conditions for local identification in endogenous single-index models.

For $q = 2, 3$, suppose that there exists a square-integrable function $\alpha_{q,\theta}$ satisfying

$$E[\alpha_{q,\theta}(Z_1) | X^T \theta] = E[Z_q | X^T \theta]. \quad (25)$$

Following the calculation for endogenous orthogonality conditions in [Ichimura and Newey \(2022\)](#), the augmented debiased moment conditions are

$$\begin{aligned} E[\{Z_2 - \alpha_{2,\theta}(Z_1)\}\{Y - h_\theta(X^T \theta)\}] &= 0, \\ E[\{Z_3 - \alpha_{3,\theta}(Z_1)\}\{Y - h_\theta(X^T \theta)\}] - \mu &= 0. \end{aligned} \quad (26)$$

3 Asymptotic Theory

This section develops the asymptotic properties of the PDG estimator. Under the stated assumptions, we establish consistency, derive the rate of convergence, and prove the oracle properties of $(\hat{\theta}, \hat{\mu})$ when the first-step function γ_0 is estimated nonparametrically. In particular, we show that

the support of $\hat{\mu}$ coincides with the support of the true auxiliary parameter μ_0 with probability approaching one. We also show that the limiting distribution of $\hat{\theta}$ and the nonzero components of $\hat{\mu}$ is the same as that of the infeasible estimator that knows the true support in advance. Building on these oracle properties, we show that, when V_n^* consistently estimates the optimal weighting matrix, $\hat{\theta}$ has the same asymptotic variance as the optimally weighted two-step GMM estimator that uses all and only the valid moments while accounting for the plug-in first step. After developing the asymptotic theory, we propose a method for selecting the tuning parameter λ_n . Finally, we present a detailed implementation procedure for the PDG estimator.

Throughout this section, $o_P(1)$ denotes convergence in probability to zero, and $O_P(1)$ denotes boundedness in probability. We slightly abuse the notation $\|\cdot\|$: when applied to a finite-dimensional vector, it denotes the standard ℓ_2 -norm, whereas, when applied to a function, it denotes a norm on the underlying function space. The intended meaning will be clear from the context. For a general vector x , let $\mathcal{J}$ be a subset of the index set of x . We use $x_{\mathcal{J}}$ to denote the subvector of x consisting only of the components indexed by $\mathcal{J}$. We write $X_n \xrightarrow{P} X$ if X_n converges to X in probability and $X_n \xrightarrow{d} X$ if X_n converges to X in distribution. Throughout, we freely reorder the coordinates of vectors and the corresponding rows and columns of conformable matrices whenever convenient. Proofs of all results in this section are collected in Appendix.

For consistency, we impose the following conditions:

Assumption 1 (Consistency). (C1) Let $g = (g_1, g_2)$. The mapping $E[g(W, \gamma_0, \theta)]$ is continuous in $\theta \in \Theta$, where Θ is compact. The true parameter θ_0 is uniquely identified by $E[g_1(W, \gamma_0, \theta)]$ in the sense that

$$E[g_1(W, \gamma_0, \theta)] = 0 \quad \text{if and only if} \quad \theta = \theta_0. \quad (27)$$

(C2) The weighting matrix satisfies $V_n^* \xrightarrow{P} V^*$, where V^* is positive definite.

(C3) For each $\ell = 1, \dots, L$,

$$\|\hat{\gamma}_\ell - \gamma_0\| = o_P(1). \quad (28)$$

(C4) For each $\ell = 1, \dots, L$ and every $\theta \in \Theta$,

$$\int \|g(w, \hat{\gamma}_\ell, \theta) - g(w, \gamma_0, \theta)\| F_0(dw) = o_P(1), \quad (29)$$

and $E[\|g(W, \gamma_0, \theta)\|] < \infty$ for all $\theta \in \Theta$.

(C5) There exist a constant $C > 0$ and a nonnegative function $d(W, \gamma)$ such that, for $\|\gamma - \gamma_0\|$ sufficiently small and all $\theta_1, \theta_2 \in \Theta$,

$$\|g(W, \gamma, \theta_2) - g(W, \gamma, \theta_1)\| \leq d(W, \gamma) \|\theta_2 - \theta_1\|^{1/C}, \quad (30)$$

where $E[d(W, \gamma)] < C$ for all such γ .

(C6) Let $\phi = (\phi_1, \phi_2)$. For each $\ell = 1, \dots, L$,

$$\int \|\phi(w, \hat{\gamma}_\ell, \alpha_0, \theta_0) - \phi(w, \gamma_0, \alpha_0, \theta_0)\|^2 F_0(dw) = o_P(1), \quad (31)$$

$$\int \|\phi(w, \gamma_0, \hat{\alpha}_\ell, \tilde{\theta}_\ell) - \phi(w, \gamma_0, \alpha_0, \theta_0)\|^2 F_0(dw) = o_P(1), \quad (32)$$

and $E[\|\phi(W, \gamma_0, \alpha_0, \theta_0)\|] < \infty$.

(C7) For each $\ell = 1, \dots, L$, let

$$\Delta_\ell(w) := \phi(w, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell) - \phi(w, \gamma_0, \hat{\alpha}_\ell, \tilde{\theta}_\ell) - \phi(w, \hat{\gamma}_\ell, \alpha_0, \theta_0) + \phi(w, \gamma_0, \alpha_0, \theta_0). \quad (33)$$

Then

$$\int \|\Delta_\ell(w)\| F_0(dw) = o_P(1). \quad (34)$$

(C8) The penalty function satisfies $\hat{P}(\mu_j) \geq 0$ for all $\mu_j \in \mathbb{R}$ and $\hat{P}(\mu_{0,j}) = o_P(1)$ for each $j = 1, \dots, m_2$.

The conditions in Assumption 1 are standard consistency and regularity conditions adapted to the present setting. C1 imposes continuity and compactness on the baseline moment conditions and ensures identification of θ_0 . C2 requires the weighting matrix to converge to a positive definite limit. C8 restricts the penalty so that its contribution at the true auxiliary parameter is asymptotically negligible, as required for consistency. Under these conditions, the consistency argument is standard once uniform convergence of the sample augmented moments to their population counterparts is established.

C3 is a high-level consistency condition on the first-step estimator $\hat{\gamma}_\ell$. Such conditions are known to hold for a broad class of commonly used machine-learning estimators under appropriate restrictions on the underlying nuisance function and the data-generating process. For example, consistency and convergence-rate results are available for the Lasso under approximate sparsity and suitable design conditions (Bickel et al., 2009), and for random forests under suitable regularity conditions (Scornet et al., 2015). More generally, Chernozhukov et al. (2018) discusses the use of a wide range of machine-learning methods for estimating nuisance functions under high-level convergence-rate conditions of this type.

Conditions C4–C7 provide the regularity conditions needed for this uniform convergence. These conditions closely parallel those imposed in Theorem A3 of Chernozhukov et al. (2022b) for consistency of the debiased GMM estimator. In particular, C4–C7 yield pointwise convergence and stochastic equicontinuity of the sample moment process, so that the results of Newey (1991) can be applied to obtain uniform convergence over Θ . An important observation is that the auxiliary parameter μ cancels from the remainder between the sample and population augmented moments. Consequently, the conditions required for uniform convergence can be stated directly in terms of the original moment functions g_1 and g_2 , without imposing additional regularity conditions with

respect to μ . In this sense, augmenting the moment conditions with the auxiliary parameter does not impose additional requirements for establishing uniform convergence.

Theorem 1 (Consistency). Let $\eta = (\theta, \mu)$, and define $\hat{\eta}$ and η_0 analogously. Under **Assumption 1**, the PDG estimator is consistent, i.e.,

$$\|\hat{\eta} - \eta_0\| = o_P(1). \quad (35)$$

To derive the rate of convergence of the PDG estimator, in addition to the assumptions imposed for consistency, we impose the following conditions:

Assumption 2 (Rate of Convergence). (R1) For $q = 1, 2$, $E[g_q(W, \gamma_0, \theta)]$ is continuously differentiable in a neighborhood of θ_0 , and

$$\frac{\partial}{\partial \theta^T} E[g_1(W, \gamma_0, \theta_0)] \quad (36)$$

has full column rank.

(R2) For each $\ell = 1, \dots, L$,

$$\int \|g(w, \hat{\gamma}_\ell, \theta_0) - g(w, \gamma_0, \theta_0)\|^2 F_0(dw) = o_P(1), \quad (37)$$

and $E[\|g(W, \gamma_0, \theta_0)\|^2] < \infty$ and $E[\|\phi(W, \gamma_0, \alpha_0, \theta_0)\|^2] < \infty$.

(R3) For each $\ell = 1, \dots, L$,

$$\int \Delta_\ell(w) F_0(dw) = o_P(1/\sqrt{n}) \quad \text{and} \quad \int \|\Delta_\ell(w)\|^2 F_0(dw) = o_P(1). \quad (38)$$

(R4) For each $\ell = 1, \dots, L$,

$$P\left(\int \phi(w, \gamma_0, \hat{\alpha}_\ell, \tilde{\theta}_\ell) F_0(dw) = 0\right) \rightarrow 1. \quad (39)$$

(R5) For each $\ell = 1, \dots, L$,

$$E[g(W, \hat{\gamma}_\ell, \theta_0) + \phi(W, \hat{\gamma}_\ell, \alpha_0, \theta_0)] - \begin{bmatrix} 0 \\ \mu_0 \end{bmatrix} = o_P(1/\sqrt{n}). \quad (40)$$

(R6) For $\|\gamma - \gamma_0\|$ sufficiently small, $g(W, \gamma, \theta)$ is continuously differentiable in θ on a neighborhood $\mathcal{N}$ of θ_0 , and there exist a constant $C > 0$ and a nonnegative function $d(W, \gamma)$ such that, for all $\theta \in \mathcal{N}$,

$$\left\| \frac{\partial}{\partial \theta^T} g(W, \gamma, \theta) - \frac{\partial}{\partial \theta^T} g(W, \gamma, \theta_0) \right\| \leq d(W, \gamma) \|\theta - \theta_0\|^{1/C}, \quad (41)$$

where $E[d(W, \gamma)] < C$ for all such γ . The Jacobian $\frac{\partial}{\partial \theta^T} g(W, \gamma_0, \theta_0)$ is integrable, differentia-

tion can be interchanged with expectation at θ_0 , and, for each $\ell = 1, \dots, L$,

$$\int \left\| \frac{\partial}{\partial \theta^T} g(w, \hat{\gamma}_\ell, \theta_0) - \frac{\partial}{\partial \theta^T} g(w, \gamma_0, \theta_0) \right\| F_0(dw) = o_P(1). \quad (42)$$

(R7) The penalty function satisfies $\hat{P}(0) = 0$. Let $\mathcal{S} = \{j \mid \mu_{0,j} \neq 0\}$ be the support of μ_0 . For each $j \in \mathcal{S}$, there exists a fixed neighborhood $\mathcal{N}_j$ of $\mu_{0,j}$ such that $0 \notin \mathcal{N}_j$, $\hat{P}(\cdot)$ is twice continuously differentiable on $\mathcal{N}_j$ with probability approaching one, and

$$\sup_{\mu \in \mathcal{N}_j} |\hat{P}''(\mu)| = o_P(1). \quad (43)$$

The conditions in Assumption 2 strengthen the conditions imposed for consistency to derive the rate of convergence of the PDG estimator. R1 imposes the usual local smoothness and rank conditions on the population moment functions. In particular, only the Jacobian of the baseline moment function $E[g_1(W, \gamma_0, \theta)]$ is required to have full column rank.

Conditions R2–R5 provide the regularity conditions needed to establish the $\sqrt{n}$ behavior of the cross-fitted debiased augmented moment at η_0 . R2 strengthens C4 at θ_0 from L_1 to mean-square convergence and imposes finite second-moment conditions. Together with C6, R2 corresponds to Assumption 1 in Chernozhukov et al. (2022a). R3 is the analogue of Assumption 2(i) in Chernozhukov et al. (2022a). Its second condition implies C7, while its first condition strengthens the population interaction remainder to the rate required for the $\sqrt{n}$ expansion. Alternative formulations of R3 are available, but this form provides a nested structure for our assumptions. Conditions R4 and R5 correspond to the recentered versions of Assumption 3(i) and (iv) in Chernozhukov et al. (2022a), where R4 is guaranteed by the properties of the debiasing term. Under C6 and R2–R5, we can derive a recentered version of Lemma 8 in Chernozhukov et al. (2022a).

R6 is a version of Assumption 5 in Chernozhukov et al. (2022a), which is used to establish convergence of the sample Jacobian matrix through a stochastic equicontinuity condition around the truth, a standard step in deriving the asymptotic distribution in this type of argument. In fact, we will use it this way when deriving the asymptotic distribution of the PDG estimator. However, it is imposed here for a different purpose: R6 ensures that the change in the stochastic remainder is $o_P(\|\hat{\theta} - \theta_0\|)$, which gives the stated convergence rate of the estimator.

Finally, R7 imposes the local regularity conditions on the penalty function needed to control the shrinkage bias on the nonzero components of μ_0 . The local uniform condition on $\hat{P}''$ slightly strengthens Assumption 3.2(iv) of Liao (2013), which imposes control only at the true parameter value. Since the penalty function may depend on n , local uniform control ensures that the second derivative remains asymptotically negligible at the intermediate point in the Taylor expansion. This condition continues to hold for the bridge, adaptive Lasso, and SCAD penalties considered in Liao (2013).

Theorem 2 (Rate of Convergence). Under **Assumptions 1 and 2**,

$$\|\hat{\eta} - \eta_0\| = O_P \left(\max \left\{ \frac{1}{\sqrt{n}}, r_n \right\} \right), \quad (44)$$

where $r_n := \max_{j \in \mathcal{S}} |\hat{P}'(\mu_{0,j})|$.

Theorem 2 shows that the convergence rate of the PDG estimator depends on two sources: the shrinkage bias induced by the penalty, which is governed by r_n , and the usual stochastic estimation error. Without the debiasing step, which adds the FSIF to the original moment function, and the use of cross-fitting, there is no general guarantee that the stochastic estimation error is of order $n^{-1/2}$ when the first-step function is estimated flexibly.

Once the stochastic estimation error is shown to attain the optimal $n^{-1/2}$ rate, selection consistency can be established by imposing additional conditions on the penalty term: the shrinkage effect of the penalty should be asymptotically negligible at the nonzero components of μ_0 while remaining sufficiently strong near zero. In particular, suppose $r_n = o_P(1/\sqrt{n})$. Then the convergence rate of the PDG estimator is governed by the stochastic estimation error and hence attains the optimal $n^{-1/2}$ rate. The condition ensuring that the penalty remains sufficiently strong near zero is more subtle. Intuitively, for each zero component of μ_0 , we require the penalized criterion to be minimized at $\hat{\mu}_j = 0$ with probability approaching one. Therefore, to establish the oracle property, we impose similar conditions as Assumption 3.3 in [Liao \(2013\)](#):

Assumption 3 (Selection Consistency). (S1) The first derivative of the penalty function on the support of μ_0 satisfies the uniform rate

$$\max_{j \in \mathcal{S}} |\hat{P}'(\mu_{0,j})| = o_P(1/\sqrt{n}). \quad (45)$$

(S2) For any $j \in \mathcal{S}^c$ and any random sequence $\{\mu_{j,n}\}_n$ such that $\mu_{j,n} = O_P(n^{-1/2})$, there exists a positive sequence a_n such that

$$\frac{a_n \lambda_n}{|\hat{P}'(\mu_{j,n})|} 1\{\mu_{j,n} \neq 0\} = O_P(1) \quad \text{and} \quad \sqrt{n} a_n \lambda_n \rightarrow \infty. \quad (46)$$

Theorem 3 (Oracle Properties). Let $\hat{\mathcal{S}}$ and $\mathcal{S}$ denote the support of $\hat{\mu}$ and μ_0 , respectively, and define $\hat{\eta}_{\mathcal{S}} := (\hat{\theta}, \hat{\mu}_{\mathcal{S}})$ and $\eta_{0,\mathcal{S}} := (\theta_0, \mu_{0,\mathcal{S}})$. Under **Assumptions 1, 2, and 3**, we have

$$\lim_{n \rightarrow \infty} P(\hat{\mathcal{S}} = \mathcal{S}) = 1. \quad (47)$$

In addition, define

$$G^* := \frac{\partial}{\partial \eta_{\mathcal{S}}^T} E[g^*(W, \gamma_0, \eta_0)]. \quad (48)$$

Then

$$\sqrt{n}(\hat{\eta}_{\mathcal{S}} - \eta_{0,\mathcal{S}}) = - \left[(G^*)^T V^* G^* \right]^{-1} (G^*)^T V^* \left[\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi^*(W_i, \gamma_0, \alpha_0, \theta_0, \mu_0) \right] + o_P(1). \quad (49)$$

3.1 Discussion on Asymptotic Efficiency

Theorem 3 provides a first-order characterization of the asymptotic behavior of the PDG estimator $\hat{\eta}$. In this subsection, we use Theorem 3 to derive the asymptotic distribution of $\hat{\theta}$, which is typically the parameter of primary interest. This distributional result provides the basis for comparing the asymptotic covariance matrix of the $\hat{\theta}$ component of the PDG estimator with that of a standard semiparametric two-step GMM estimator. We first present the covariance matrix under a general weighting matrix and then specialize it to optimal weighting, following [Newey \(1994\)](#). After formally stating the result, we discuss how this efficiency comparison should be interpreted and highlight several important qualifications and potential pitfalls in interpreting the result.

Let

$$g_o(W, \gamma, \theta) := \begin{bmatrix} g_1(W, \gamma, \theta) \\ g_{2,\mathcal{S}^c}(W, \gamma, \theta) \end{bmatrix} \quad (50)$$

denote the oracle moment function obtained by stacking the baseline moments g_1 and the components of the additional moment function g_2 indexed by $\mathcal{S}^c$, which correspond to the zero components of μ_0 . Thus, g_o contains all valid moments within the candidate collection (g_1, g_2) . Because g_o contains the baseline moment function g_1 , the identification condition imposed on g_1 implies that, given γ_0, θ_0 is the unique solution to

$$E[g_o(W, \gamma_0, \theta_0)] = 0. \quad (51)$$

Suppose that a consistent estimator $\hat{\gamma}$ of the nonparametric component γ_0 is available for constructing a two-step plug-in GMM estimator of θ_0 . Under suitable problem-specific regularity conditions, if the weighting matrix satisfies $V_n \xrightarrow{P} V$, where V is positive definite, the asymptotic covariance matrix of the standard semiparametric two-step GMM estimator is

$$\left[G^T V G \right]^{-1} G^T V E \left[\{g_o + \phi_o\} \{g_o + \phi_o\}^T \right] V G \left[G^T V G \right]^{-1}, \quad (52)$$

where

$$G := \frac{\partial}{\partial \theta^T} E[g_o(W, \gamma_0, \theta_0)] \quad (53)$$

is the Jacobian of the oracle population moment function with respect to θ , evaluated at the truth, and

$$\phi_o(W, \gamma, \alpha, \theta) := \begin{bmatrix} \phi_1(W, \gamma, \alpha, \theta) \\ \phi_{2,\mathcal{S}^c}(W, \gamma, \alpha, \theta) \end{bmatrix} \quad (54)$$

is the corresponding oracle FSIF obtained by retaining the components of ϕ associated with the valid oracle moments. The inclusion of ϕ_o accounts for the first-order effect of estimating the

nonparametric component γ_0 , as in [Newey \(1994\)](#).

Define

$$\Omega := E \left[\{g_o + \phi_o\} \{g_o + \phi_o\}^T \right], \quad (55)$$

where g_o and ϕ_o are evaluated at the truth. Suppose that Ω is positive definite. By choosing the weighting matrix V_n such that

$$V_n \xrightarrow{p} V = \Omega^{-1}, \quad (56)$$

the asymptotic covariance matrix of the two-step GMM estimator reduces to

$$\left[G^T \Omega^{-1} G \right]^{-1}. \quad (57)$$

By the standard GMM efficiency argument, this covariance matrix is the smallest among the class of two-step plug-in GMM estimators indexed by the weighting matrix. Thus, the optimally weighted oracle two-step GMM estimator provides a natural benchmark for evaluating the efficiency of the PDG estimator.

For the augmented moment system, define

$$\Omega^* := E \left[\psi^* (\psi^*)^T \right], \quad (58)$$

where ψ^* is evaluated at the truth. The following theorem shows that, when the PDG estimator is also optimally weighted, the asymptotic covariance matrix of $\hat{\theta}$ coincides with that of the optimally weighted oracle two-step GMM estimator.

Theorem 4. Under **Assumptions 1, 2, and 3**, suppose that Ω^* is positive definite and

$$V_n^* \xrightarrow{p} V^* = (\Omega^*)^{-1}. \quad (59)$$

Then

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N \left(0, \left[G^T \Omega^{-1} G \right]^{-1} \right), \quad (60)$$

where G and Ω are defined above.

Theorem 4 shows that the asymptotic covariance matrix of $\hat{\theta}$ under the PDG estimator coincides with that of the optimally weighted oracle two-step GMM estimator described above. However, two qualifications are important for interpreting this result correctly.

First, like other sparse estimators possessing an oracle property, the PDG estimator is nonregular over models that allow the support of μ_0 to change under local perturbations of order $n^{-1/2}$. Accordingly, Theorems 3 and 4 are pointwise asymptotic results under a fixed data-generating process and do not provide a uniform approximation over sequences of data-generating processes in which some components of μ_0 approach zero as the sample size increases. Under such local alternatives, valid and slightly invalid moments may be difficult to distinguish. This is the familiar post-model-selection inference problem emphasized by [Liao \(2013\)](#) and studied in detail by [Leeb](#)

and Pötscher (2005).

Nevertheless, this qualification does not diminish the practical role of the PDG estimator. The procedure provides a data-driven benchmark that jointly estimates θ_0 and determines which candidate moment conditions should be treated as valid, rather than requiring the researcher to specify the valid set in advance. Thus, although the oracle approximation does not by itself resolve the general post-model-selection inference problem, the PDG estimator provides a systematic framework for conducting moment selection and parameter estimation simultaneously (Liao, 2013).

Second, when the nonparametric component γ_0 is known, the ordinary optimally weighted GMM estimator has the smallest asymptotic covariance matrix in the class of weighted GMM estimators based on the same moment conditions, in the sense that the difference between the asymptotic covariance matrix under any other choice of weighting matrix and that under optimal weighting is always positive semidefinite. In addition, when the data are i.i.d., the covariance matrix of the optimally weighted GMM estimator is also the semiparametric efficiency bound among all regular estimators in the corresponding fully specified semiparametric model (Chamberlain, 1987). Related results show that optimally weighted semiparametric two-step GMM estimators can also attain the efficiency bound when the model identifying the unknown nuisance functions is fully specified (Akerberg et al., 2014). Without further specifying the underlying probability model and the structure imposed on γ_0 , the semiparametric efficiency bound is unclear in the current context.

Moreover, suppose we specialize the current setting to that of Akerberg et al. (2014). Because of the irregularity of the PDG estimator discussed above, it is not in the class of estimators considered when deriving the semiparametric efficiency bound. Therefore, comparing its pointwise asymptotic covariance matrix with the semiparametric efficiency bound would not establish the semiparametric efficiency of the estimator. The results derived in Theorem 4 should therefore be viewed as a benchmark for comparison, rather than as establishing that the PDG estimator is semiparametrically efficient.

3.2 Implementation Details

In this subsection, we discuss implementation details of the estimator, focusing on the adaptive Lasso penalty. The advantages of the adaptive Lasso are twofold. First, its large-sample behavior is intuitive from its construction. As the sample size increases, the adaptive weight remains bounded for a nonzero component $\mu_{0,j}$ but diverges for a zero component $\mu_{0,j}$. Second, conditional on the adaptive weights, the adaptive Lasso penalty is convex, in contrast to many other commonly used oracle penalties, which are nonconvex. This provides an important computational advantage by avoiding the additional nonconvexity introduced by the penalty.

We first show that the adaptive Lasso penalty satisfies all restrictions on the penalty function imposed in Assumptions 1–3, so that the results established in the previous sections apply when this penalty is used. We then propose a practical choice of the tuning parameter λ_n in $\hat{P}_{\lambda_n}(\mu_j)$, together with coordinate-specific loadings constructed from the cross-fitted debiased moments. For a discussion of other penalty functions for moment selection, see Appendix B of Liao (2013).

Proposition 1 (Adaptive Lasso Penalty). Let $\hat{P}(\mu_j)$ be the adaptive Lasso penalty defined previously, and suppose $\tilde{\mu}$ is a $\sqrt{n}$ -consistent initial estimator of the auxiliary parameter μ . In addition, if the tuning parameter λ_n satisfies $\sqrt{n}\lambda_n \rightarrow 0$ and $n^{(1+r)/2}\lambda_n \rightarrow \infty$, then the adaptive Lasso penalty satisfies C8, R7, S1, and S2, where the scaling sequence in S2 can be taken to be $a_n = n^{r/2}$.

Proposition 1 requires only that the initial estimator $\tilde{\mu}$ be $\sqrt{n}$ -consistent, so different constructions are available. However, since the construction of the cross-fitted debiased moments already requires the fold-specific preliminary estimators $\tilde{\theta}_\ell$, $\hat{\gamma}_\ell$, and $\hat{\alpha}_\ell$, we propose a construction of $\tilde{\mu}$ that requires no additional preliminary estimation. In particular, define

$$\tilde{\mu} := \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \left[g_2(W_i, \hat{\gamma}_\ell, \tilde{\theta}_\ell) + \phi_2(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell) \right]. \quad (61)$$

Thus, no additional preliminary estimator is required beyond those already used to construct the cross-fitted debiased moments. The following proposition shows that the $\tilde{\mu}$ constructed above is indeed $\sqrt{n}$ -consistent.

Proposition 2 (Initial Estimator of μ_0). Under C3, C6, and R2–R6, suppose that for each $\ell = 1, \dots, L$, $\tilde{\theta}_\ell$ is $\sqrt{n}$ -consistent. Then $\tilde{\mu}$ is a $\sqrt{n}$ -consistent estimator of μ_0 .

We note that this proposition is directly related to the $\sqrt{n}$ -estimability of the population mean of moment functions with a first-step nonparametric component, which is a key feature of the debiased moments. Such a result need not hold in general if $\tilde{\mu}$ is instead defined as the cross-fitted average of the original moment functions without adding the FSIF adjustment.

Next, guided by the rate conditions in Proposition 1, we propose a feasible calibration of the adaptive Lasso penalty that accounts for the finite-sample behavior of the PDG estimator. We focus on the case where $r = 2$ to simplify the notation and introduce coordinate-specific loading terms to account for scale differences across the moment conditions. Although the theoretical analysis uses a common tuning parameter across components, we treat the coordinate-specific loadings as finite-sample adjustments that do not alter the asymptotic order of the penalty. This approach is algebraically equivalent to standardizing the moment functions, which is a common implementation technique in the Lasso literature.

Suppose that $\lambda_n = 2cn^{-\kappa}$ for some fixed constant $c > 0$ and exponent κ . The conditions in Proposition 1 require

$$\frac{1}{2} < \kappa < \frac{3}{2}. \quad (62)$$

We choose κ to be the midpoint of this interval, which gives

$$\lambda_n = 2cn^{-1}. \quad (63)$$

The choice of c does not affect whether the asymptotic rate conditions in Proposition 1 are satisfied. However, we would like to choose c to account for the finite-sample behavior of the estimator and

ensure that the penalty is invariant to the scale of the moment conditions. To achieve the latter objective, we replace the common constant c with coordinate-specific loading terms.

For $i \in I_\ell$, define

$$\hat{\psi}_i := g(W_i, \hat{\gamma}_\ell, \tilde{\theta}_\ell) + \phi(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell), \quad \bar{\psi} := \frac{1}{n} \sum_{i=1}^n \hat{\psi}_i, \quad (64)$$

and construct

$$\hat{\Omega}^* := \frac{1}{n} \sum_{i=1}^n (\hat{\psi}_i - \bar{\psi})(\hat{\psi}_i - \bar{\psi})^T. \quad (65)$$

The centering in this expression accounts for the nonzero mean of the candidate moment block. For each candidate moment j , define the coordinate-specific loading

$$\hat{c}_j := \left([\hat{\Omega}^*]_{m_1+j, m_1+j}^T e_{m_1+j}^T V_n^* \hat{\Omega}^* V_n^* e_{m_1+j} \right)^{1/2}, \quad (66)$$

where e_{m_1+j} denotes the (m_1+j) th coordinate vector. We then use the coordinate-specific adaptive Lasso penalty

$$\hat{P}(\mu_j) = 2\hat{c}_j n^{-1} \frac{|\mu_j|}{|\tilde{\mu}_j|^2}. \quad (67)$$

To understand the motivation for the proposed choice of $\hat{c}_j$, note that the loading should account for the variability of the j th score in the estimation problem while ensuring that the penalty is invariant to rescaling of the moment conditions. Recall that, when $\hat{\mu}_j \neq 0$, the first-order condition implies

$$\left| e_{m_1+j}^T V_n^* \sqrt{n} \bar{\psi}^*(\hat{\eta}) \right| = \frac{\hat{c}_j n^{-1/2}}{|\tilde{\mu}_j|^2}. \quad (68)$$

The quantity

$$\left(e_{m_1+j}^T V_n^* \hat{\Omega}^* V_n^* e_{m_1+j} \right)^{1/2} \quad (69)$$

is the estimated standard deviation of the j th KKT score, while

$$[\hat{\Omega}^*]_{m_1+j, m_1+j}^{1/2} \quad (70)$$

is the estimated standard deviation of the j th candidate moment. Because $r = 2$, the square of the latter quantity enters $\hat{c}_j$. Together, these two components ensure that the first-order condition compares a standardized score with the preliminary estimate $\tilde{\mu}_j$ measured relative to the scale of the corresponding candidate moment.

Given the proposed coordinate-specific loadings, the recommended estimation procedure for the optimally weighted PDG estimator with an adaptive Lasso penalty for $r = 1$ is described in Algorithm 1. For more detailed implementation guidance, see Appendix B.

Algorithm 1 Estimation Procedure for the PDG Estimator

Step 1: Sample splitting. Partition the sample into $L \geq 3$ disjoint folds $I_1, \dots, I_L$.

Step 2: Inner preliminary estimation. For each $\ell = 1, \dots, L$, exclude I_ℓ and use the remaining $L - 1$ folds as the inner cross-fitting partition. For each $\ell' \neq \ell$, use only observations outside $I_\ell \cup I_{\ell'}$ to construct the first-step nuisance estimators and a consistent starting estimator required to evaluate the baseline debiased moments on $I_{\ell'}$, following Chernozhukov et al. (2022a).

Step 3: Baseline-only debiased GMM iteration. Evaluate the baseline debiased moments $g_1 + \phi_1$ on each $I_{\ell'}$, aggregate them over $\ell' \neq \ell$, and minimize the corresponding baseline-only GMM criterion with a general symmetric positive definite weighting matrix to obtain $\tilde{\theta}_\ell$. The candidate moments g_2 are not used in this step.

Step 4: Outer first-step estimation. Using all observations in I_ℓ^c , construct $\hat{\gamma}_\ell$ and $\hat{\alpha}_\ell$, using $\tilde{\theta}_\ell$ when required. These estimators are used to evaluate observations in I_ℓ .

Step 5: Adaptive penalty factors. Using the fold-specific estimators constructed above, evaluate the preliminary cross-fitted debiased moments and construct $\tilde{\mu}$ as defined above. Use $1/|\tilde{\mu}_j|^2$ as the adaptive penalty factor for the j th candidate moment.

Step 6: Optimal weighting matrix. Construct $\hat{\Omega}^*$ from the preliminary cross-fitted debiased moments and set $V_n^* = (\hat{\Omega}^*)^{-1}$.

Step 7: Coordinate-specific loading terms. For each $j = 1, \dots, m_2$, construct

$$\hat{c}_j = \left([\hat{\Omega}^*]_{m_1+j, m_1+j}^2 [V_n^*]_{m_1+j, m_1+j} \right)^{1/2}, \quad (71)$$

and define the corresponding adaptive Lasso penalty as

$$\hat{P}(\mu_j) = 2\hat{c}_j n^{-1} \frac{|\mu_j|}{|\tilde{\mu}_j|^2}. \quad (72)$$

Step 8: PDG estimation. Construct $\bar{\psi}^*(\theta, \mu)$ using the fold-specific estimators obtained above and obtain $(\hat{\theta}, \hat{\mu})$ by minimizing the PDG criterion with weighting matrix V_n^* and the coordinate-specific adaptive Lasso penalties constructed in Step 7.

3.3 Inference and Post-Selection Estimation

For inference, let $\hat{\mathcal{S}}$ denote the support of $\hat{\mu}$, which estimates the support of μ_0 . Let $\hat{\Omega}$ denote the principal submatrix of $\hat{\Omega}^*$ corresponding to the baseline moments g_1 and the candidate moments $g_{2,\hat{\mathcal{S}}^c}$. Define

$$\hat{G} := \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \frac{\partial}{\partial \theta^T} \begin{bmatrix} g_1(W_i, \hat{\gamma}_\ell, \hat{\theta}) \\ g_{2,\hat{\mathcal{S}}^c}(W_i, \hat{\gamma}_\ell, \hat{\theta}) \end{bmatrix}. \quad (73)$$

A plug-in estimator of the asymptotic covariance matrix of $\sqrt{n}(\hat{\theta} - \theta_0)$ under optimal weighting is

$$\left[\hat{G}^T \hat{\Omega}^{-1} \hat{G} \right]^{-1}. \quad (74)$$

As discussed previously, in finite samples, naive confidence intervals constructed using this plug-in estimator do not account for moment-selection errors. Uniformly valid post-selection inference is beyond the scope of the current paper. Under correct moment selection, the plug-in estimator is consistent under the additional condition that, for each $\ell = 1, \dots, L$,

$$\int \left\| g(w, \hat{\gamma}_\ell, \tilde{\theta}_\ell) - g(w, \hat{\gamma}_\ell, \theta_0) \right\|^2 F_0(dw) = o_P(1). \quad (75)$$

The argument is similar to the proof of Lemma E1 in [Chernozhukov et al. \(2022b\)](#), so we omit the details.

Alternatively, in view of the selection consistency property established in Theorem 3, we recommend the post-selection estimator as the final reported estimator in practice. Given the PDG estimator $(\hat{\theta}, \hat{\mu})$ and the estimated support $\hat{\mathcal{S}}$, define the post-selection PDG estimator, denoted by $\hat{\theta}_{\text{post}}$, as the unpenalized debiased GMM estimator of [Chernozhukov et al. \(2022a\)](#), with the feasible optimal weight matrix V_n , using only the baseline moments g_1 and the candidate moments $g_{2,\hat{\mathcal{S}}^c}$. It uses the same sample splitting, nuisance estimates, and first-step influence-function adjustments as the PDG estimator.

As the following proposition shows, $\hat{\theta}_{\text{post}}$ and $\hat{\theta}$ have the same asymptotic distribution under optimal weighting. In finite samples, however, $\hat{\theta}_{\text{post}}$ avoids the direct shrinkage bias induced by the penalty in the final estimation step. This can substantially improve finite-sample performance, as illustrated by our simulation results.

Proposition 3 (Post-Selection PDG). Under **Assumptions 1, 2, and 3**, suppose that Ω^* is positive definite and

$$V_n^* \xrightarrow{p} V^* = (\Omega^*)^{-1}. \quad (76)$$

Then

$$\sqrt{n}(\hat{\theta}_{\text{post}} - \theta_0) \xrightarrow{d} N \left(0, \left[G^T \Omega^{-1} G \right]^{-1} \right), \quad (77)$$

where G and Ω are defined above.

4 Simulations

In this section, we report the simulation results for the PDG estimator with an adaptive Lasso penalty for $r = 2$, following the estimation procedure described in Algorithm 1 and using the partially linear IV model described in Example 1. Consider the following data-generating process:

$$Y_i = X_i\theta_0 + h_0(U_i) + \epsilon_i, \quad (78)$$

where $\theta_0 = 0.5$ is the parameter of interest and $U_i = (U_{i,1}, \dots, U_{i,10})$ has independent $\text{Unif}[-1, 1]$ coordinates. Define

$$b(U) = [\sin(\pi U_1), U_2^2 - 1/3, U_3, \cos(\pi U_4), U_5^3]^T, \quad (79)$$

so that only the first five controls enter the true conditional means. The conditional mean of X and the nonparametric component of Y are

$$r_0(U) = (0.6, 0.5, 0.4, 0.3, 0.2)b(U) \quad \text{and} \quad h_0(U) = (0.7, -0.6, 0.5, 0.4, -0.3)b(U). \quad (80)$$

Consequently,

$$\ell_0(U) := E[Y | U] = \theta_0 r_0(U) + h_0(U). \quad (81)$$

Next, let

$$a^{(1)} = (0.60, 0.25, 0.10, 0.20, 0.05)^T, \quad a^{(2)} = (0.05, 0.60, 0.25, 0.10, 0.20)^T, \quad (82)$$

$$a^{(3)} = (0.20, 0.05, 0.60, 0.25, 0.10)^T, \quad a^{(4)} = (0.10, 0.20, 0.05, 0.60, 0.25)^T, \quad (83)$$

$$a^{(5)} = (0.25, 0.10, 0.20, 0.05, 0.60)^T. \quad (84)$$

Define

$$m_{1,0}(U) = a^{(1)T}b(U), \quad m_{2,j,0}(U) = a^{(1+(j \bmod 5))^T}b(U), \quad j = 1, \dots, 25. \quad (85)$$

Independently of U_i , draw

$$A_i = (A_{0i}, \dots, A_{5i})^T \sim N(0, \Sigma_A), \quad [\Sigma_A]_{jk} = 0.2^{|j-k|}, \quad (86)$$

for $j, k = 0, \dots, 5$, together with

$$F_i \sim N(0, I_{20}), \quad \begin{pmatrix} \epsilon_i \\ \nu_i \end{pmatrix} \sim N\left(0, \begin{pmatrix} 1 & 0.6 \\ 0.6 & 1 \end{pmatrix}\right). \quad (87)$$

The blocks U_i , A_i , F_i , and (ϵ_i, ν_i) are mutually independent. The endogenous variable is generated by

$$X_i = r_0(U_i) + \pi_1 A_{0i} + (0.15, 0.15, 0.20, 0.20, 0.25)(A_{1i}, \dots, A_{5i})^T + 0.05 \sum_{k=1}^{20} F_{ki} + \nu_i, \quad (88)$$

where $\pi_1 \in \{0.2, 0.8\}$ controls the relevance of the maintained-valid baseline instrument. Note that $r_0(U) = E[X | U]$ and that endogeneity follows from the correlation between ϵ_i and ν_i .

The baseline instrument is

$$Z_{1i} = m_{1,0}(U_i) + A_{0i}. \quad (89)$$

The first five candidates,

$$Z_{2,j,i} = m_{2,j,0}(U_i) + A_{ji}, \quad j = 1, \dots, 5, \quad (90)$$

generate valid additional moments. For $k = 1, \dots, 20$, define

$$\delta_k = c_{\text{inv}} + (0.8 - c_{\text{inv}}) \frac{k-1}{19}, \quad c_{\text{inv}} \in \{0.2, 0.5\}, \quad (91)$$

and generate the remaining candidates as

$$Z_{2,5+k,i} = m_{2,5+k,0}(U_i) + F_{ki} + \delta_k \epsilon_i. \quad (92)$$

These twenty candidate moments have nonzero means δ_k at the true parameter. The parameter c_{inv} controls the degree of invalidity by setting the smallest moment violation among the twenty invalid candidate instruments. Thus,

$$\mu_0 = (0, 0, 0, 0, 0, \delta_1, \dots, \delta_{20})^T, \quad \mathcal{S} = \{6, \dots, 25\}, \quad \mathcal{S}^c = \{1, \dots, 5\}, \quad (93)$$

where $\mathcal{S}$ is the invalid support and $\mathcal{S}^c$ is the valid candidate set.

The four designs are

$$\begin{aligned} A: & (\pi_1, c_{\text{inv}}) = (0.2, 0.2), & B: & (\pi_1, c_{\text{inv}}) = (0.2, 0.5), \\ C: & (\pi_1, c_{\text{inv}}) = (0.8, 0.2), & D: & (\pi_1, c_{\text{inv}}) = (0.8, 0.5). \end{aligned} \quad (94)$$

Each design is evaluated at $n = 500, 1000, 2000$, giving twelve cells, with $B = 10,000$ independent replications per cell. All five estimators completed every replication, giving 120,000 datasets and 600,000 estimates.

The nuisance functions ℓ_0 , r_0 , m_1 , and $m_{2,1}, \dots, m_{2,25}$ are estimated using an additive cubic B-spline ridge estimator with all ten controls. This estimator is chosen for computational efficiency.

We report five estimators. Oracle-valid GMM uses Z_1 and all five valid candidate instruments, whereas Baseline-only GMM uses Z_1 alone. Both estimators use the FSIF adjustment and five-fold

cross-fitting, following the locally robust estimation framework of Chernozhukov et al. (2022a). PDG is the proposed estimator in this paper, using an adaptive Lasso penalty with $r = 2$ and the penalty loadings discussed in the previous section. Post-PDG reuses PDG’s first-step nuisance estimates, fold assignments, and FSIF correction, but refits θ by unpenalized GMM using Z_1 and the candidate moments selected as valid by PDG. Post-PDG is our recommended estimator for practical applications. Finally, NPG is a naive penalized GMM benchmark that uses full-sample nuisance estimates without the FSIF adjustment or cross-fitting.

Table 1 reports the empirical probabilities of unsafe, exact, and conservative moment selection for PDG and NPG across the four designs and three sample sizes. Selection is safe when $\hat{\mathcal{S}}^c \subseteq \mathcal{S}^c$, so that no invalid candidate moment is retained. Within this category, selection is exact when $\hat{\mathcal{S}}^c = \mathcal{S}^c$ and conservative when $\hat{\mathcal{S}}^c \subsetneq \mathcal{S}^c$, meaning that at least one valid candidate moment is omitted. Unsafe selection occurs when $\hat{\mathcal{S}}^c \not\subseteq \mathcal{S}^c$, so that at least one invalid candidate moment is retained. The baseline moment is always included. Tables 2–5 report the bias, empirical standard deviation, and root mean squared error of the five estimators for Designs A–D, respectively, at $n = 500, 1000, 2000$. Table 6 reports the average estimated standard error and the empirical coverage probability of the nominal 95% confidence interval for each estimator and cell; coverage is the fraction of replications in which the interval contains θ_0 . Figure 1 presents kernel density estimates of the simulated distributions of $\hat{\theta}$ for Oracle-valid GMM, Baseline-only GMM, PDG, and Post-PDG at $n = 1000$. Its four panels correspond to Designs A–D, and the vertical reference line marks the true value $\theta_0 = 0.5$.

The tables demonstrate substantial precision gains from exploiting valid additional moments. At $n = 1000$ and $n = 2000$, PDG achieves high moment-selection accuracy, and Post-PDG reduces empirical variance relative to Baseline-only GMM by approximately 81.75–82.77% when $\pi_1 = 0.2$ and 26.33–27.68% when $\pi_1 = 0.8$. These reductions closely match the corresponding population Oracle variance reductions of 81.71% and 26.49%. At these sample sizes, Post-PDG combines these precision gains with small bias, RMSE close to the Oracle benchmark, and coverage between 92.34% and 94.81%. Direct PDG, however, exhibits substantial shrinkage bias despite its lower dispersion. The $n = 500$ experiments provide a more demanding assessment, with more frequent selection errors and lower coverage in some designs; extreme estimates also make variance comparisons less stable in Design B. Nevertheless, Post-PDG has lower absolute bias, lower RMSE, and coverage closer to the nominal level than direct PDG in every cell.

We note that such pronounced shrinkage bias reflects the continued penalization of nonzero auxiliary coefficients under our construction of the loading term $\hat{c}_j$. In Liao (2013), the loading is constructed from a preliminary shrinkage fit. For moments classified as invalid in that fit, the corresponding loadings equal zero, so their auxiliary coefficients are no longer penalized in the subsequent estimation. In contrast, our loading is strictly positive and does not depend on a preliminary support estimate. Post-PDG removes this continuing penalization by refitting the selected valid moments without a penalty, and its simulated densities resemble the corresponding oracle densities.

Table 1: Moment-selection probabilities

Design (π_1, c_{inv})	$n = 500$ (U, E, C)	$n = 1000$ (U, E, C)	$n = 2000$ (U, E, C)
<i>PDG</i>			
A (0.2, 0.2)	(0.6716, 0.2971, 0.0313)	(0.0150, 0.9295, 0.0555)	(0.0000, 0.9795, 0.0205)
B (0.2, 0.5)	(0.0330, 0.8528, 0.1142)	(0.0001, 0.9504, 0.0495)	(0.0000, 0.9826, 0.0174)
C (0.8, 0.2)	(0.3124, 0.6685, 0.0191)	(0.0062, 0.9843, 0.0095)	(0.0000, 0.9973, 0.0027)
D (0.8, 0.5)	(0.0000, 0.9716, 0.0284)	(0.0000, 0.9885, 0.0115)	(0.0000, 0.9981, 0.0019)
<i>NPG</i>			
A (0.2, 0.2)	(0.8295, 0.1637, 0.0068)	(0.0224, 0.9455, 0.0321)	(0.0000, 0.9887, 0.0113)
B (0.2, 0.5)	(0.0235, 0.9139, 0.0626)	(0.0000, 0.9718, 0.0282)	(0.0000, 0.9922, 0.0078)
C (0.8, 0.2)	(0.5389, 0.4581, 0.0030)	(0.0108, 0.9872, 0.0020)	(0.0000, 0.9988, 0.0012)
D (0.8, 0.5)	(0.0001, 0.9941, 0.0058)	(0.0000, 0.9972, 0.0028)	(0.0000, 0.9996, 0.0004)

Notes: Each cell reports (U, E, C): unsafe, exact, and conservative selection probabilities. Unsafe retains at least one invalid candidate; exact retains all and only the five valid candidates; conservative retains a strict subset of the valid candidates and no invalid candidates. Each panel uses 10,000 completed replications per design and sample size. Probabilities sum to one before rounding. The maximum selection-probability Monte Carlo standard error is 0.0050.

Table 2: Estimation performance: Design A, $(\pi_1, c_{\text{inv}}) = (0.2, 0.2)$

Estimator	$n = 500$			$n = 1000$			$n = 2000$		
	Bias	SD	RMSE	Bias	SD	RMSE	Bias	SD	RMSE
Oracle	0.0264	0.1410	0.1435	0.0119	0.0619	0.0630	0.0054	0.0420	0.0423
PDG	0.4316	0.1471	0.4560	0.1906	0.0579	0.1992	0.0883	0.0380	0.0962
Post-PDG	0.1480	0.2038	0.2518	0.0158	0.0650	0.0669	0.0060	0.0423	0.0427
Baseline	-0.0259	0.4717	0.4723	-0.0123	0.1521	0.1526	-0.0042	0.1005	0.1006
NPG	0.4346	0.1119	0.4488	0.2170	0.0578	0.2245	0.1055	0.0373	0.1119

Notes: SD is the empirical standard deviation with divisor $B - 1$; RMSE is the square root of mean squared error relative to $\theta_0 = 0.5$. All estimators completed $B = 10,000$ replications per sample size. The maximum Monte Carlo standard error across reported bias and RMSE entries is 4.8806.

Table 3: Estimation performance: Design B, $(\pi_1, c_{\text{inv}}) = (0.2, 0.5)$

Estimator	$n = 500$			$n = 1000$			$n = 2000$		
	Bias	SD	RMSE	Bias	SD	RMSE	Bias	SD	RMSE
Oracle	0.0261	1.0712	1.0715	0.0109	0.0617	0.0626	0.0049	0.0417	0.0420
PDG	0.3656	0.8621	0.9364	0.1504	0.0555	0.1603	0.0701	0.0381	0.0798
Post-PDG	0.0602	0.8430	0.8451	0.0135	0.0636	0.0650	0.0054	0.0420	0.0424
Baseline	-0.1598	11.6290	11.6296	-0.0091	0.1515	0.1518	-0.0057	0.1013	0.1014
NPG	0.4063	0.1166	0.4227	0.1885	0.0536	0.1960	0.0922	0.0369	0.0993

Notes: SD is the empirical standard deviation with divisor $B - 1$; RMSE is the square root of mean squared error relative to $\theta_0 = 0.5$. All estimators completed $B = 10,000$ replications per sample size. The maximum Monte Carlo standard error across reported bias and RMSE entries is 4.8806.

Table 4: Estimation performance: Design C, $(\pi_1, c_{\text{inv}}) = (0.8, 0.2)$

Estimator	$n = 500$			$n = 1000$			$n = 2000$		
	Bias	SD	RMSE	Bias	SD	RMSE	Bias	SD	RMSE
Oracle	0.0080	0.0521	0.0527	0.0034	0.0346	0.0348	0.0013	0.0239	0.0239
PDG	0.1298	0.0507	0.1393	0.0592	0.0330	0.0678	0.0279	0.0232	0.0363
Post-PDG	0.0160	0.0543	0.0566	0.0035	0.0347	0.0349	0.0013	0.0239	0.0239
Baseline	-0.0012	0.0612	0.0612	-0.0006	0.0407	0.0407	-0.0006	0.0280	0.0280
NPG	0.1186	0.0551	0.1308	0.0579	0.0338	0.0671	0.0288	0.0232	0.0370

Notes: SD is the empirical standard deviation with divisor $B - 1$; RMSE is the square root of mean squared error relative to $\theta_0 = 0.5$. All estimators completed $B = 10,000$ replications per sample size. The maximum Monte Carlo standard error across reported bias and RMSE entries is 4.8806.

Table 5: Estimation performance: Design D, $(\pi_1, c_{\text{inv}}) = (0.8, 0.5)$

Estimator	$n = 500$			$n = 1000$			$n = 2000$		
	Bias	SD	RMSE	Bias	SD	RMSE	Bias	SD	RMSE
Oracle	0.0080	0.0525	0.0531	0.0037	0.0342	0.0344	0.0018	0.0239	0.0240
PDG	0.1036	0.0500	0.1150	0.0472	0.0326	0.0574	0.0227	0.0232	0.0325
Post-PDG	0.0080	0.0526	0.0532	0.0037	0.0343	0.0345	0.0018	0.0239	0.0240
Baseline	-0.0018	0.0612	0.0612	-0.0005	0.0404	0.0404	-0.0002	0.0279	0.0279
NPG	0.1147	0.0520	0.1259	0.0547	0.0328	0.0638	0.0275	0.0232	0.0360

Notes: SD is the empirical standard deviation with divisor $B - 1$; RMSE is the square root of mean squared error relative to $\theta_0 = 0.5$. All estimators completed $B = 10,000$ replications per sample size. The maximum Monte Carlo standard error across reported bias and RMSE entries is 4.8806.

Table 6: Standard errors and nominal 95% coverage

Estimator	$n = 500$		$n = 1000$		$n = 2000$	
	Avg. SE	Coverage	Avg. SE	Coverage	Avg. SE	Coverage
$A: (\pi_1, c_{\text{inv}}) = (0.2, 0.2)$						
Oracle	0.1022	0.9243	0.0617	0.9342	0.0419	0.9433
PDG	0.0953	0.0076	0.0619	0.1215	0.0419	0.4335
Post-PDG	0.0953	0.6350	0.0619	0.9234	0.0419	0.9406
Baseline	0.2745	0.9517	0.1493	0.9534	0.0995	0.9516
NPG	0.0941	0.0074	0.0687	0.0870	0.0481	0.3870
$B: (\pi_1, c_{\text{inv}}) = (0.2, 0.5)$						
Oracle	0.1240	0.9314	0.0616	0.9371	0.0419	0.9459
PDG	0.1142	0.0446	0.0618	0.3026	0.0420	0.6093
Post-PDG	0.1142	0.8911	0.0618	0.9295	0.0420	0.9436
Baseline	0.4487	0.9568	0.1490	0.9520	0.0996	0.9514
NPG	0.0994	0.0127	0.0686	0.1683	0.0482	0.5169
$C: (\pi_1, c_{\text{inv}}) = (0.8, 0.2)$						
Oracle	0.0521	0.9404	0.0345	0.9439	0.0238	0.9477
PDG	0.0519	0.2997	0.0345	0.5895	0.0238	0.7766
Post-PDG	0.0519	0.9196	0.0345	0.9430	0.0238	0.9477
Baseline	0.0621	0.9544	0.0407	0.9487	0.0279	0.9500
NPG	0.0551	0.4209	0.0386	0.6837	0.0271	0.8381
$D: (\pi_1, c_{\text{inv}}) = (0.8, 0.5)$						
Oracle	0.0521	0.9391	0.0345	0.9468	0.0238	0.9482
PDG	0.0521	0.4862	0.0345	0.7155	0.0238	0.8319
Post-PDG	0.0521	0.9388	0.0345	0.9461	0.0238	0.9481
Baseline	0.0621	0.9514	0.0407	0.9501	0.0279	0.9493
NPG	0.0554	0.4522	0.0385	0.7220	0.0271	0.8496

Notes: Avg. SE is the mean estimated standard error; it is not empirical SD. Coverage is an unconditional probability, so 0.0076 means 0.76% coverage of a nominal 95% interval. Each entry uses 10,000 completed replications. All intervals are centered at their own estimator. NPG uses naive plug-in inference without nuisance correction. The maximum coverage Monte Carlo standard error is 0.0050.

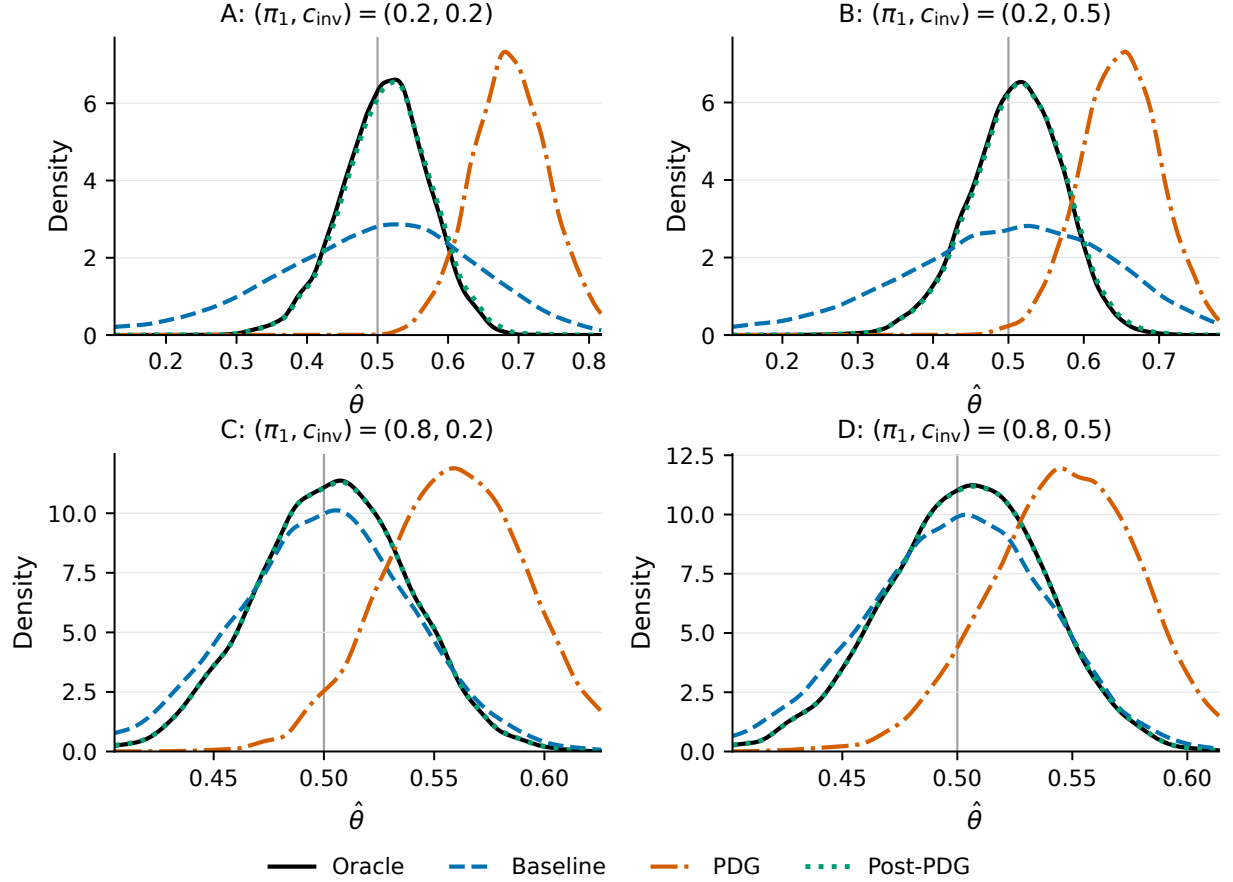


Figure 1: Sampling densities at $n = 1000$

Notes: Gaussian kernel densities of raw $\hat{\theta}$ based on 10,000 saved estimates per estimator and panel; the vertical line marks $\theta_0 = 0.5$. Each estimator uses the robust Silverman bandwidth $0.9 \min\{s, \text{IQR} / 1.34\} B^{-1/5}$. Within each panel, all four curves share a 512-point grid over the pooled 0.5th–99.5th percentiles. All observations enter the bandwidth and density calculations, including those outside the display bounds. The largest estimator-specific fraction outside its panel's display is 3.32%; all individual fractions are supplied with the figure data in the replication package. Colors and line styles are both varied to distinguish curves.

Table 7: Variance gains from valid additional moments

Design	n	ns_{Baseline}^2	Reduction vs. Baseline (%)		Variance / Oracle	
			PDG	Post-PDG	PDG	Post-PDG
A	500	111.2296	90.27	81.33	1.0886	2.0878
A	1000	23.1321	85.52	81.75	0.8745	1.1018
A	2000	20.2067	85.69	82.27	0.8202	1.0159
B	500	67617.3915	99.45	99.47	0.6477	0.6193
B	1000	22.9565	86.57	82.40	0.8104	1.0625
B	2000	20.5168	85.83	82.77	0.8364	1.0173
C	500	1.8719	31.34	21.25	0.9478	1.0871
C	1000	1.6575	34.40	27.43	0.9078	1.0044
C	2000	1.5626	31.40	26.89	0.9392	1.0009
D	500	1.8730	33.39	26.18	0.9063	1.0045
D	1000	1.6283	34.90	27.68	0.9042	1.0045
D	2000	1.5563	30.59	26.33	0.9437	1.0016

Notes: s_e^2 is empirical variance across 10,000 completed replications, calculated from unrounded estimates with divisor $B - 1$. The reductions are $100(1 - s_e^2/s_{\text{Baseline}}^2)$; the last two columns are $s_e^2/s_{\text{Oracle}}^2$. Multiplying Baseline variance by n permits comparison with its root- n asymptotic variance. Designs A–D have $(\pi_1, c_{\text{inv}}) = (0.2, 0.2), (0.2, 0.5), (0.8, 0.2), (0.8, 0.5)$, respectively. A ratio below one measures lower empirical dispersion, not a bias-adjusted efficiency gain. No variance-comparison significance test is imposed.

5 Conclusion

This paper establishes that the oracle properties of penalized moment selection are preserved in the presence of nonparametric or machine learning first-step plug-in estimators. By combining orthogonalization and cross-fitting with penalization, the proposed PDG procedure consistently identifies the invalid moment set and, under optimal weighting, has the same asymptotic distribution as the oracle debiased GMM estimator that knows the valid moment set in advance. These results hold under the stated conditions for a fixed data-generating process. The post-selection estimator inherits this oracle distribution. Thus, the procedure retains the precision benefits of additional valid moments despite uncertainty about moment validity and the presence of estimated first-step components, with asymptotic variance no larger than that of baseline-only estimation.

The simulations show that PDG accurately selects the moment set but exhibits substantial shrinkage bias in the direct penalized estimate. Post-PDG substantially reduces this bias, and its simulated distributions closely resemble the corresponding oracle distributions.

Several extensions are left for future research. First, our oracle results are pointwise and do not establish uniformly valid inference when invalid moment means approach zero. Second, the analysis maintains a fixed number of moments, leaving open an extension to high-dimensional settings in which the number of candidate moments grows with the sample size. Finally, equivalence to the optimally weighted oracle debiased GMM estimator does not establish semiparametric efficiency. Characterizing the semiparametric efficiency bound and determining conditions under which the proposed procedure attains it remain important questions for future work.

A Proofs of Main Results

Lemma 1. Let

$$R_n(\theta) := \bar{\psi}^*(\eta) - E[g^*(W, \gamma_0, \eta)] = \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \left\{ g(W_i, \hat{\gamma}_\ell, \theta) + \phi(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell) \right\} - E[g(W, \gamma_0, \theta)]. \quad (95)$$

Under C1 and C3–C7,

$$\sup_{\theta \in \Theta} \|R_n(\theta)\| = o_P(1). \quad (96)$$

Proof. Note that μ cancels in the expression, so we can rewrite it as

$$\begin{aligned} R_n(\theta) &= \frac{1}{n} \sum_{i=1}^n [g(W_i, \gamma_0, \theta) - E[g(W, \gamma_0, \theta)] + \phi(W_i, \gamma_0, \alpha_0, \theta_0)] \\ &\quad + \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} [g(W_i, \hat{\gamma}_\ell, \theta) - g(W_i, \gamma_0, \theta)] \\ &\quad + \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} [\phi(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell) - \phi(W_i, \gamma_0, \alpha_0, \theta_0)]. \end{aligned} \quad (97)$$

The first term in the decomposition is $o_P(1)$ by the LLN and $E[\phi(W, \gamma_0, \alpha_0, \theta_0)] = 0$.

For the second term, let $\mathcal{W}_\ell^c$ denote the observations in I_ℓ^c . Then

$$\begin{aligned} &P \left(\left\| \frac{1}{n} \sum_{i \in I_\ell} [g(W_i, \hat{\gamma}_\ell, \theta) - g(W_i, \gamma_0, \theta)] \right\| > \epsilon \right) \\ &= E \left[P \left(\left\| \frac{1}{n} \sum_{i \in I_\ell} [g(W_i, \hat{\gamma}_\ell, \theta) - g(W_i, \gamma_0, \theta)] \right\| > \epsilon \mid \mathcal{W}_\ell^c \right) \right], \end{aligned} \quad (98)$$

and, by the conditional Markov inequality, the expression inside the expectation is bounded by

$$\min \left\{ 1, \frac{1}{\epsilon} \int \|g(w, \hat{\gamma}_\ell, \theta) - g(w, \gamma_0, \theta)\| F_0(dw) \right\}, \quad (99)$$

where the latter integral is $o_P(1)$ by C4, whence the expectation is $o(1)$.

For the third term, note that

$$\begin{aligned} &\phi(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell) - \phi(W_i, \gamma_0, \alpha_0, \theta_0) \\ &= \phi(W_i, \hat{\gamma}_\ell, \alpha_0, \theta_0) - \phi(W_i, \gamma_0, \alpha_0, \theta_0) + \phi(W_i, \gamma_0, \hat{\alpha}_\ell, \tilde{\theta}_\ell) - \phi(W_i, \gamma_0, \alpha_0, \theta_0) + \Delta_\ell(W_i). \end{aligned} \quad (100)$$

By the triangle inequality and Jensen's inequality,

$$\begin{aligned}
& \int \left\| \phi(w, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell) - \phi(w, \gamma_0, \alpha_0, \theta_0) \right\| F_0(dw) \\
& \leq \left[\int \left\| \phi(w, \hat{\gamma}_\ell, \alpha_0, \theta_0) - \phi(w, \gamma_0, \alpha_0, \theta_0) \right\|^2 F_0(dw) \right]^{1/2} \\
& \quad + \left[\int \left\| \phi(w, \gamma_0, \hat{\alpha}_\ell, \tilde{\theta}_\ell) - \phi(w, \gamma_0, \alpha_0, \theta_0) \right\|^2 F_0(dw) \right]^{1/2} \\
& \quad + \int \|\Delta_\ell(w)\| F_0(dw).
\end{aligned} \tag{101}$$

By Assumptions C6 and C7, this term is $o_P(1)$. Then

$$\begin{aligned}
& P \left(\left\| \frac{1}{n} \sum_{i \in I_\ell} [\phi(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell) - \phi(W_i, \gamma_0, \alpha_0, \theta_0)] \right\| > \epsilon \right) \\
& = E \left[P \left(\left\| \frac{1}{n} \sum_{i \in I_\ell} [\phi(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell) - \phi(W_i, \gamma_0, \alpha_0, \theta_0)] \right\| > \epsilon \mid \mathcal{W}_\ell^c \right) \right]
\end{aligned} \tag{102}$$

is $o(1)$ by a similar argument as for the second term.

Together, we have shown that for each fixed θ , $\|R_n(\theta)\| = o_P(1)$.

For $\theta_1, \theta_2 \in \Theta$, by C3, we invoke C5 to obtain

$$\begin{aligned}
\|R_n(\theta_2) - R_n(\theta_1)\| & \leq \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \|g(W_i, \hat{\gamma}_\ell, \theta_2) - g(W_i, \hat{\gamma}_\ell, \theta_1)\| \\
& \quad + E[\|g(W, \gamma_0, \theta_2) - g(W, \gamma_0, \theta_1)\|] \\
& \leq \left[\frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} d(W_i, \hat{\gamma}_\ell) + E[d(W, \gamma_0)] \right] \|\theta_2 - \theta_1\|^{1/C}.
\end{aligned} \tag{103}$$

The Lipschitz term in the bracket is $O_P(1)$, since for $M > 0$, by the conditional Markov inequality,

$$\begin{aligned}
P \left(\frac{1}{n} \sum_{i \in I_\ell} d(W_i, \hat{\gamma}_\ell) > M \right) & = E \left[P \left(\frac{1}{n} \sum_{i \in I_\ell} d(W_i, \hat{\gamma}_\ell) > M \mid \mathcal{W}_\ell^c \right) \right] \\
& \leq E \left[\frac{1}{M} E[d(W, \hat{\gamma}_\ell)] \right] \\
& \leq \frac{C}{M}.
\end{aligned} \tag{104}$$

Therefore, $R_n(\theta)$ satisfies Assumption 3A in Newey (1991), whence

$$\sup_{\theta \in \Theta} \|R_n(\theta)\| = o_P(1) \tag{105}$$

by Corollary 2.2 in Newey (1991). □

Proof of Theorem 1. Next, note that

$$\bar{\psi}^*(\eta_0) = \bar{\psi}^*(\eta_0) - E[g^*(W, \gamma_0, \eta_0)] = R_n(\theta_0), \quad (106)$$

whence

$$\bar{\psi}^*(\hat{\eta})^T V_n^* \bar{\psi}^*(\hat{\eta}) \leq R_n(\theta_0)^T V_n^* R_n(\theta_0) + \sum_{j=1}^{m_2} \hat{P}(\mu_{0,j}), \quad (107)$$

by the definition of $\hat{\eta}$. Write

$$\bar{\psi}^*(\hat{\eta}) = E[g^*(W, \gamma_0, \hat{\eta})] + \bar{\psi}^*(\hat{\eta}) - E[g^*(W, \gamma_0, \hat{\eta})], \quad (108)$$

then

$$\begin{aligned} \bar{\psi}^*(\hat{\eta})^T V_n^* \bar{\psi}^*(\hat{\eta}) &\geq \frac{1}{2} E[g^*(W, \gamma_0, \hat{\eta})]^T V_n^* E[g^*(W, \gamma_0, \hat{\eta})] \\ &\quad - \left[\bar{\psi}^*(\hat{\eta}) - E[g^*(W, \gamma_0, \hat{\eta})] \right]^T V_n^* \left[\bar{\psi}^*(\hat{\eta}) - E[g^*(W, \gamma_0, \hat{\eta})] \right] \\ &\geq \frac{1}{2} E[g^*(W, \gamma_0, \hat{\eta})]^T V_n^* E[g^*(W, \gamma_0, \hat{\eta})] - \sup_{\theta \in \Theta} R_n(\theta)^T V_n^* R_n(\theta), \end{aligned} \quad (109)$$

where the inequality follows from $(x + y)^T A (x + y) \geq \frac{1}{2} x^T A x - y^T A y$ for any positive semidefinite matrix A and conformable vectors x and y .

Combining the above results and using C8, we have

$$E[g^*(W, \gamma_0, \hat{\eta})]^T V_n^* E[g^*(W, \gamma_0, \hat{\eta})] \leq 4 \sup_{\theta \in \Theta} R_n(\theta)^T V_n^* R_n(\theta) + 2 \sum_{j=1}^{m_2} \hat{P}(\mu_{0,j}) = o_P(1), \quad (110)$$

whence

$$\|E[g^*(W, \gamma_0, \hat{\eta})]\| = o_P(1). \quad (111)$$

By C1, $\hat{\theta}$ is consistent, since for any $\epsilon > 0$, there exists a $\delta > 0$ such that

$$P\left(\|\hat{\theta} - \theta_0\| > \epsilon\right) \leq P\left(\|E[g_1(W, \gamma_0, \hat{\theta})]\| > \delta\right), \quad (112)$$

which converges to zero. In addition, $\hat{\mu}$ is consistent, since

$$\|E[g_2(W, \gamma_0, \hat{\theta})] - \hat{\mu}\| = \|\mu_0 + o_P(1) - \hat{\mu}\| = o_P(1). \quad (113)$$

□

Lemma 2. Under C6 and R2–R5,

$$\sqrt{n} \bar{\psi}^*(\eta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n [g^*(W_i, \gamma_0, \theta_0, \mu_0) + \phi(W_i, \gamma_0, \alpha_0, \theta_0)] + o_P(1). \quad (114)$$

Proof. The argument is identical to the proof of Lemma 8 in Chernozhukov et al. (2022a), except that we recenter it at the true population expectation. Write

$$\begin{aligned}
\sqrt{n}\bar{\psi}^*(\eta_0) &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \left[g(W_i, \gamma_0, \theta_0) + \phi(W_i, \gamma_0, \alpha_0, \theta_0) - \begin{bmatrix} 0 \\ \mu_0 \end{bmatrix} \right] \\
&\quad + \frac{1}{\sqrt{n}} \sum_{\ell=1}^L \sum_{i \in I_\ell} \left[g(W_i, \hat{\gamma}_\ell, \theta_0) - g(W_i, \gamma_0, \theta_0) \right. \\
&\quad \quad + \phi(W_i, \hat{\gamma}_\ell, \alpha_0, \theta_0) - \phi(W_i, \gamma_0, \alpha_0, \theta_0) \\
&\quad \quad \left. + \phi(W_i, \gamma_0, \hat{\alpha}_\ell, \tilde{\theta}_\ell) - \phi(W_i, \gamma_0, \alpha_0, \theta_0) \right] \\
&\quad + \frac{1}{\sqrt{n}} \sum_{\ell=1}^L \sum_{i \in I_\ell} \Delta_\ell(W_i),
\end{aligned} \tag{115}$$

where the first term is $O_P(1)$ and the last term is $o_P(1)$ by R3. It remains to show that the second term of the decomposition is $o_P(1)$.

Denote the summand in the second term of the decomposition by $S_{\ell i}$. Conditional on $\mathcal{W}_\ell^c$, the conditional expectation of $S_{\ell i}$ is

$$E[g(W, \hat{\gamma}_\ell, \theta_0) + \phi(W, \hat{\gamma}_\ell, \alpha_0, \theta_0)] - \begin{bmatrix} 0 \\ \mu_0 \end{bmatrix} + E[\phi(W, \gamma_0, \hat{\alpha}_\ell, \tilde{\theta}_\ell)] = o_P(1/\sqrt{n}) \tag{116}$$

by R4 and R5, whence

$$\frac{1}{\sqrt{n}} \sum_{i \in I_\ell} E[S_{\ell i} \mid \mathcal{W}_\ell^c] = o_P(1). \tag{117}$$

Using the inequality $\|a + b + c\|^2 \leq 3(\|a\|^2 + \|b\|^2 + \|c\|^2)$, by R2 and C6,

$$E[\|S_{\ell i}\|^2 \mid \mathcal{W}_\ell^c] = o_P(1). \tag{118}$$

Together,

$$\frac{1}{\sqrt{n}} \sum_{i \in I_\ell} S_{\ell i} = \frac{1}{\sqrt{n}} \sum_{i \in I_\ell} (S_{\ell i} - E[S_{\ell i} \mid \mathcal{W}_\ell^c]) + \frac{1}{\sqrt{n}} \sum_{i \in I_\ell} E[S_{\ell i} \mid \mathcal{W}_\ell^c] = o_P(1). \tag{119}$$

as in equation (A.6) of Chernozhukov et al. (2022a). $\square$

Lemma 3. Under the conditions for consistency, C3, and R6,

$$\|R_n(\hat{\theta}) - R_n(\theta_0)\| = o_P(\|\hat{\theta} - \theta_0\|). \tag{120}$$

Proof. By consistency and C3, eventually $\hat{\gamma}_\ell$ enters the neighborhood where R6 applies, and $\hat{\theta} \in \mathcal{N}$. By the continuous differentiability of $g(W, \gamma, \theta)$ on the set $\mathcal{N}$, the Fundamental Theorem of Calculus

implies

$$\begin{aligned}
g(W_i, \hat{\gamma}_\ell, \hat{\theta}) - g(W_i, \hat{\gamma}_\ell, \theta_0) &= \left[\int_0^1 \frac{\partial}{\partial \theta^T} g(W_i, \hat{\gamma}_\ell, \theta_0 + t(\hat{\theta} - \theta_0)) dt \right] (\hat{\theta} - \theta_0) \\
&= \frac{\partial}{\partial \theta^T} g(W_i, \hat{\gamma}_\ell, \theta_0) (\hat{\theta} - \theta_0) \\
&\quad + \left[\int_0^1 \left(\frac{\partial}{\partial \theta^T} g(W_i, \hat{\gamma}_\ell, \theta_0 + t(\hat{\theta} - \theta_0)) - \frac{\partial}{\partial \theta^T} g(W_i, \hat{\gamma}_\ell, \theta_0) \right) dt \right] (\hat{\theta} - \theta_0).
\end{aligned} \tag{121}$$

Denote the second term by $T_{\ell i}$. By the Lipschitz condition on the Jacobian imposed by R6, we have

$$\|T_{\ell i}\| \leq \int_0^1 d(W_i, \hat{\gamma}_\ell) \|t(\hat{\theta} - \theta_0)\|^{1/C} dt \|\hat{\theta} - \theta_0\| = \frac{C}{1+C} d(W_i, \hat{\gamma}_\ell) \|\hat{\theta} - \theta_0\|^{1+1/C}. \tag{122}$$

Applying the same argument at γ_0 and then taking expectations yields

$$E[g(W, \gamma_0, \hat{\theta})] - E[g(W, \gamma_0, \theta_0)] = E \left[\frac{\partial}{\partial \theta^T} g(W, \gamma_0, \theta_0) \right] (\hat{\theta} - \theta_0) + T_0, \tag{123}$$

where, by Fubini's theorem,

$$T_0 = \left[\int_0^1 E \left[\frac{\partial}{\partial \theta^T} g(W, \gamma_0, \theta_0 + t(\hat{\theta} - \theta_0)) - \frac{\partial}{\partial \theta^T} g(W, \gamma_0, \theta_0) \right] dt \right] (\hat{\theta} - \theta_0). \tag{124}$$

By the same Lipschitz condition in R6,

$$\begin{aligned}
\|T_0\| &\leq \int_0^1 E \left[\left\| \frac{\partial}{\partial \theta^T} g(W, \gamma_0, \theta_0 + t(\hat{\theta} - \theta_0)) - \frac{\partial}{\partial \theta^T} g(W, \gamma_0, \theta_0) \right\| \right] dt \|\hat{\theta} - \theta_0\| \\
&\leq \frac{C}{1+C} E[d(W, \gamma_0)] \|\hat{\theta} - \theta_0\|^{1+1/C}.
\end{aligned} \tag{125}$$

Using the above expansions, we have

$$\begin{aligned}
R_n(\hat{\theta}) - R_n(\theta_0) &= \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} [g(W_i, \hat{\gamma}_\ell, \hat{\theta}) - g(W_i, \hat{\gamma}_\ell, \theta_0)] - E [g(W, \gamma_0, \hat{\theta}) - g(W, \gamma_0, \theta_0)] \\
&= \left[\frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \frac{\partial}{\partial \theta^T} g(W_i, \hat{\gamma}_\ell, \theta_0) - E \left[\frac{\partial}{\partial \theta^T} g(W, \gamma_0, \theta_0) \right] \right] (\hat{\theta} - \theta_0) \\
&\quad + \left[\frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} T_{\ell i} - T_0 \right] \\
&= \left[\frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \left(\frac{\partial}{\partial \theta^T} g(W_i, \hat{\gamma}_\ell, \theta_0) - \frac{\partial}{\partial \theta^T} g(W_i, \gamma_0, \theta_0) \right) \right] (\hat{\theta} - \theta_0) \\
&\quad + \left[\frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta^T} g(W_i, \gamma_0, \theta_0) - E \left[\frac{\partial}{\partial \theta^T} g(W, \gamma_0, \theta_0) \right] \right] (\hat{\theta} - \theta_0) \\
&\quad + \left[\frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} T_{\ell i} - T_0 \right].
\end{aligned} \tag{126}$$

The coefficient of the first $(\hat{\theta} - \theta_0)$ is $o_P(1)$ by the L_1 convergence condition on the Jacobian in R6 and the conditional Markov inequality, as used several times in the preceding proofs. The coefficient of the second $(\hat{\theta} - \theta_0)$ is $o_P(1)$ by the LLN, which applies by the integrability of the Jacobian imposed in R6. Using the bounds on the remainders established above,

$$\left\| \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} T_{\ell i} - T_0 \right\| \leq \left[\frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} d(W_i, \hat{\gamma}_\ell) + E[d(W, \gamma_0)] \right] \|\hat{\theta} - \theta_0\|^{1+1/C}, \quad (127)$$

where the term in brackets is $O_P(1)$ by the same argument used in Lemma 1, whence the remainder term is $o_P(\|\hat{\theta} - \theta_0\|)$ by the consistency of $\hat{\theta}$. $\square$

Proof of Theorem 2. By Lemmas 2 and 3,

$$R_n(\hat{\theta}) = R_n(\theta_0) + R_n(\hat{\theta}) - R_n(\theta_0) = O_P(1/\sqrt{n}) + o_P(\|\hat{\theta} - \theta_0\|), \quad (128)$$

whence, by C2,

$$R_n(\hat{\theta})^T V_n^* R_n(\hat{\theta}) = O_P(1/n) + o_P(\|\hat{\theta} - \theta_0\|^2). \quad (129)$$

By the twice continuous differentiability of $\hat{P}(\mu_j)$ on $\mathcal{N}_j$ imposed by R7, a first-order Taylor expansion with Lagrange remainder gives

$$\begin{aligned} \left| \sum_{j \in \mathcal{S}} [\hat{P}(\hat{\mu}_j) - \hat{P}(\mu_{0,j})] \right| &\leq \max_{j \in \mathcal{S}} |\hat{P}'(\mu_{0,j})| \sum_{j \in \mathcal{S}} |\hat{\mu}_j - \mu_{0,j}| + \frac{1}{2} \max_{j \in \mathcal{S}} \sup_{\mu_j \in \mathcal{N}_j} |\hat{P}''(\mu_j)| \sum_{j \in \mathcal{S}} (\hat{\mu}_j - \mu_{0,j})^2 \\ &\leq c_1 r_n \|\hat{\mu} - \mu_0\| + o_P(\|\hat{\mu} - \mu_0\|^2), \end{aligned} \quad (130)$$

where the intermediate point in the Taylor expansion eventually lies in $\mathcal{N}_j$ by the consistency of $\hat{\mu}$, and $c_1 = \sqrt{|\mathcal{S}|}$.

By the differentiability of $E[g^*(W, \gamma_0, \eta)]$ at η_0 , which is implied by R1 and the definition of g^* ,

$$\begin{aligned} &E[g^*(W, \gamma_0, \hat{\eta})]^T V_n^* E[g^*(W, \gamma_0, \hat{\eta})] \\ &= (\hat{\eta} - \eta_0)^T \left[\frac{\partial}{\partial \eta^T} E[g^*(W, \gamma_0, \eta_0)] \right]^T V_n^* \frac{\partial}{\partial \eta^T} E[g^*(W, \gamma_0, \eta_0)] (\hat{\eta} - \eta_0) + o_P(\|\hat{\eta} - \eta_0\|^2) \\ &\geq c_2 \|\hat{\eta} - \eta_0\|^2 + o_P(\|\hat{\eta} - \eta_0\|^2) \\ &= (c_2 + o_P(1)) \|\hat{\eta} - \eta_0\|^2, \end{aligned} \quad (131)$$

where c_2 is some positive constant that exists by C2 and R1.

Recall from the proof of Theorem 1 that we established the bound

$$\bar{\psi}^*(\hat{\eta})^T V_n^* \bar{\psi}^*(\hat{\eta}) \geq \frac{1}{2} E[g^*(W, \gamma_0, \hat{\eta})]^T V_n^* E[g^*(W, \gamma_0, \hat{\eta})] - R_n(\hat{\theta})^T V_n^* R_n(\hat{\theta}). \quad (132)$$

By the conditions on the penalty and the definition of $\hat{\eta}$, we have

$$\bar{\psi}^*(\hat{\eta})^T V_n^* \bar{\psi}^*(\hat{\eta}) + \sum_{j \in \mathcal{S}} \left[\hat{P}(\hat{\mu}_j) - \hat{P}(\mu_{0,j}) \right] \leq \bar{\psi}^*(\eta_0)^T V_n^* \bar{\psi}^*(\eta_0). \quad (133)$$

Using the above inequalities and all the previously established bounds, we obtain

$$\begin{aligned} & (c_2 + o_P(1)) \|\hat{\eta} - \eta_0\|^2 - 2c_1 r_n \|\hat{\mu} - \mu_0\| - o_P(\|\hat{\mu} - \mu_0\|^2) \\ & \leq 2\bar{\psi}^*(\eta_0)^T V_n^* \bar{\psi}^*(\eta_0) + 2R_n(\hat{\theta})^T V_n^* R_n(\hat{\theta}) \\ & = O_P(1/n) + O_P(1/n) + o_P(\|\hat{\theta} - \theta_0\|^2). \end{aligned} \quad (134)$$

Note that $\|\hat{\mu} - \mu_0\| \leq \|\hat{\eta} - \eta_0\|$. After absorbing the terms of the same order, we have

$$(c_2 + o_P(1)) \|\hat{\eta} - \eta_0\|^2 \leq O_P(1/n) + 2c_1 r_n \|\hat{\eta} - \eta_0\|. \quad (135)$$

Since

$$P\left(c_2 + o_P(1) \geq \frac{c_2}{2}\right) \rightarrow 1, \quad (136)$$

it suffices to prove the result on the event where $c_2 + o_P(1) \geq c_2/2$. Therefore,

$$\begin{aligned} \|\hat{\eta} - \eta_0\|^2 & \leq O_P(1/n) + \frac{4c_1}{c_2} r_n \|\hat{\eta} - \eta_0\| \\ & \leq O_P(1/n) + \frac{8c_1^2}{c_2^2} r_n^2 + \frac{1}{2} \|\hat{\eta} - \eta_0\|^2, \end{aligned} \quad (137)$$

whence

$$\|\hat{\eta} - \eta_0\|^2 = O_P(1/n + r_n^2). \quad (138)$$

By taking square roots,

$$\|\hat{\eta} - \eta_0\| = O_P(1/\sqrt{n}) + O_P(r_n) = O_P\left(\max\left\{\frac{1}{\sqrt{n}}, r_n\right\}\right). \quad (139)$$

□

Lemma 4. Under C3 and R6, if $\|\bar{\eta} - \eta_0\| = o_P(1)$, then

$$\frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \frac{\partial}{\partial \eta^T} g^*(W_i, \hat{\gamma}_\ell, \bar{\eta}) \rightarrow \frac{\partial}{\partial \eta^T} E[g^*(W, \gamma_0, \eta_0)] \quad (140)$$

in probability.

Proof. The proof follows from an argument similar to the proof of Lemma E2 in [Chernozhukov et al. \(2022b\)](#) and the fact that the Jacobian of g^* depends on η only through θ . □

Proof of Theorem 3. We first prove selection consistency. For $j \in \mathcal{S}^c$, assume that $\hat{\mu}_j \neq 0$. The

first-order condition implies that

$$2 \left[\frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \frac{\partial}{\partial \mu_j} g^*(W_i, \hat{\gamma}_\ell, \hat{\theta}, \hat{\mu}) \right] V_n^* \bar{\psi}^*(\hat{\theta}, \hat{\mu}) + \hat{P}'(\hat{\mu}_j) = 0, \quad (141)$$

where $\frac{\partial}{\partial \mu_j} g^*(W_i, \hat{\gamma}_\ell, \hat{\theta}, \hat{\mu})$ is the j th row of the matrix

$$\frac{\partial}{\partial \mu} g^*(W_i, \hat{\gamma}_\ell, \hat{\theta}, \hat{\mu}) = \begin{bmatrix} O_{m_2 \times m_1} & -I_{m_2 \times m_2} \end{bmatrix}. \quad (142)$$

Expanding $\bar{\psi}^*(\hat{\theta}, \hat{\mu})$ at (θ_0, μ_0) implies that

$$\left| [V_n^*]_{(m_1+j)\text{th row}} \left(\sqrt{n} \bar{\psi}^*(\theta_0, \mu_0) + \left[\frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \frac{\partial}{\partial \eta^T} g^*(W_i, \hat{\gamma}_\ell, \bar{\eta}) \right] \sqrt{n}(\hat{\eta} - \eta_0) \right) \right| = \left| \frac{\sqrt{n}}{2} \hat{P}'(\hat{\mu}_j) \right|, \quad (143)$$

where $\bar{\eta}$ is an intermediate point between $\hat{\eta}$ and η_0 , which may differ across components of g^* . By C2, S1, Theorem 2, and Lemmas 2 and 4, every term on the LHS of the above expression is $O_P(1)$; hence, the LHS of the above equality is $O_P(1)$. Denote it by A_n .

On the other hand, since $j \in \mathcal{S}^c$, on the event where $\hat{\mu}_j \neq 0$, Theorem 2 implies that $\hat{\mu}_j = O_P(n^{-1/2})$. Denote the right-hand side by B_n . By S2,

$$\frac{1}{B_n} = \frac{2}{\sqrt{n} |\hat{P}'(\hat{\mu}_j)|} = \frac{2}{\sqrt{n} a_n \lambda_n} \frac{a_n \lambda_n}{|\hat{P}'(\hat{\mu}_j)|} = o_P(1). \quad (144)$$

For $j \in \mathcal{S}^c$, let $\epsilon > 0$. There exists $M > 0$ such that

$$P(\hat{\mu}_j \neq 0) \leq P(A_n = B_n) \leq P(|A_n| > M) + P(|B_n| \leq M) \leq \epsilon + P(1/|B_n| \geq 1/M), \quad (145)$$

where the second term converges to zero, whence

$$P(\mathcal{S}^c \subseteq \hat{\mathcal{S}}^c) \rightarrow 1. \quad (146)$$

The reverse containment follows from the consistency.

For the asymptotic linear expansion, it suffices to derive the result assuming $\hat{\eta} = (\hat{\eta}_S, \hat{\mu}_{S^c})$, where $\hat{\mu}_{S^c} = 0$, whence the first-order condition on the nonzero components implies

$$2 \left[\frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \frac{\partial}{\partial \eta_S^T} g^*(W_i, \hat{\gamma}_\ell, \hat{\eta}_S, 0) \right]^T V_n^* \bar{\psi}^*(\hat{\eta}_S, 0) + \sum_{j \in \mathcal{S}} \frac{\partial}{\partial \eta_S} \hat{P}(\hat{\mu}_j) = 0. \quad (147)$$

Let

$$\bar{G}^*(\eta) := \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \frac{\partial}{\partial \eta_S^T} g^*(W_i, \hat{\gamma}_\ell, \eta). \quad (148)$$

Expanding $\bar{\psi}^*(\hat{\eta}_{\mathcal{S}}, 0)$ at $\eta_{0,\mathcal{S}}$ and solving for $\sqrt{n}(\hat{\eta}_{\mathcal{S}} - \eta_{0,\mathcal{S}})$, we have

$$\begin{aligned} \sqrt{n}(\hat{\eta}_{\mathcal{S}} - \eta_{0,\mathcal{S}}) = & - \left[\bar{G}^*(\hat{\eta}_{\mathcal{S}}, 0)^T V_n^* \bar{G}^*(\bar{\eta}_{\mathcal{S}}, 0) \right]^{-1} \bar{G}^*(\hat{\eta}_{\mathcal{S}}, 0)^T V_n^* \sqrt{n} \bar{\psi}^*(\eta_0) \\ & - \left[\bar{G}^*(\hat{\eta}_{\mathcal{S}}, 0)^T V_n^* \bar{G}^*(\bar{\eta}_{\mathcal{S}}, 0) \right]^{-1} \frac{\sqrt{n}}{2} \sum_{j \in \mathcal{S}} \frac{\partial}{\partial \eta_{\mathcal{S}}} \hat{P}(\hat{\mu}_j), \end{aligned} \quad (149)$$

where $\bar{\eta}_{\mathcal{S}}$ is an intermediate point, which may differ across components of g^* . Then the result follows from C2, R1, R7, S1, Theorem 2, Lemmas 2 and 4, and Slutsky's lemma. $\square$

Proof of Theorem 4. By Theorem 3 and the condition $V^* = (\Omega^*)^{-1}$, the asymptotic covariance matrix of $\hat{\eta}_{\mathcal{S}}$ is

$$\left[(G^*)^T (\Omega^*)^{-1} G^* \right]^{-1}, \quad (150)$$

where

$$G^* = \begin{bmatrix} G & O \\ \frac{\partial}{\partial \theta^T} E[g_{2,\mathcal{S}}(W, \gamma_0, \theta_0)] & -I \end{bmatrix}, \quad (151)$$

and the upper-left block of Ω^* is Ω .

By construction, the asymptotic covariance matrix of $\hat{\theta}$ is the upper-left $d_{\theta} \times d_{\theta}$ submatrix of the asymptotic covariance matrix of $\hat{\eta}_{\mathcal{S}}$. Then the result follows from the formula for the inverse of a partitioned matrix and an application of the continuous mapping theorem. $\square$

Proof of Proposition 1. C8 holds because for $j \in \mathcal{S}^c$, $\hat{P}(\mu_{0,j}) = 0$, and for $j \in \mathcal{S}$, $\hat{P}(\mu_{0,j}) = \lambda_n O_P(1) = o_P(1)$. R7 holds trivially since $\hat{P}''(\mu) = 0$ for all $\mu \neq 0$.

For S1, let $j \in \mathcal{S}$. Since $\tilde{\mu}_j$ is consistent,

$$\sqrt{n}|\hat{P}'(\mu_{0,j})| = \sqrt{n}\lambda_n O_P(1) = o_P(1). \quad (152)$$

Lastly, for S2, take $a_n = n^{r/2}$. First, note that $\sqrt{n}a_n\lambda_n = n^{(1+r)/2}\lambda_n \rightarrow \infty$. Let $j \in \mathcal{S}^c$ and let $\mu_{j,n} = O_P(1/\sqrt{n})$ be any random sequence. For $\mu_{j,n} \neq 0$,

$$|\hat{P}'(\mu_{j,n})| = \frac{\lambda_n}{|\tilde{\mu}_j|^r}. \quad (153)$$

Then

$$\frac{a_n\lambda_n}{|\hat{P}'(\mu_{j,n})|} 1\{\mu_{j,n} \neq 0\} = |\sqrt{n}\tilde{\mu}_j|^r 1\{\mu_{j,n} \neq 0\} = O_P(1). \quad (154)$$

$\square$

Proof of Proposition 2. Note that

$$\begin{aligned}\tilde{\mu} - \mu_0 &= \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \left[g_2(W_i, \hat{\gamma}_\ell, \theta_0) + \phi_2(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell) - \mu_0 \right] \\ &\quad + \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \left[g_2(W_i, \hat{\gamma}_\ell, \tilde{\theta}_\ell) - g_2(W_i, \hat{\gamma}_\ell, \theta_0) \right].\end{aligned}\tag{155}$$

The first term is $O_P(1/\sqrt{n})$ by Lemma 2. The second term is also $O_P(1/\sqrt{n})$ by an argument similar to that used in Lemma 3 and the assumption that $\tilde{\theta}_\ell - \theta_0 = O_P(1/\sqrt{n})$. $\square$

Proof of Proposition 3. Let $\hat{\theta}_o$ denote the debiased GMM estimator as in Chernozhukov et al. (2022a), using the baseline moments g_1 and the oracle valid candidate moments g_{2,S^c} , with the same nuisance estimates, FSIF adjustment, and optimal weighting rule as $\hat{\theta}_{\text{post}}$. Under the current assumptions, $\sqrt{n}(\hat{\theta}_o - \theta_0)$ has the asymptotic distribution stated in the proposition. By construction, for every $\epsilon > 0$,

$$P\left(\sqrt{n}\|\hat{\theta}_{\text{post}} - \hat{\theta}_o\| > \epsilon\right) \leq P(\hat{\mathcal{S}} \neq \mathcal{S}) \rightarrow 0,\tag{156}$$

whence

$$\sqrt{n}(\hat{\theta}_{\text{post}} - \theta_0) = \sqrt{n}(\hat{\theta}_o - \theta_0) + o_P(1).\tag{157}$$

$\square$

B Implementation Guide

This appendix provides a detailed walkthrough of Algorithm 1 with $L = 5$ and the adaptive Lasso penalty with $r = 2$. Partition the sample into five disjoint folds $I_1, \dots, I_5$ of approximately equal size.

Inner first-step estimation. For each outer fold I_ℓ , use the remaining four folds as the inner cross-fitting partition. For each $\ell' \neq \ell$, define

$$T_{\ell,\ell'} := \{1, \dots, n\} \setminus (I_\ell \cup I_{\ell'}).\tag{158}$$

Using only observations in $T_{\ell,\ell'}$, construct the inner nuisance estimator $\hat{\gamma}_{\ell,\ell'}$ and the starting estimator $\check{\theta}_{\ell,\ell'}$. One option for the starting estimator is the baseline plug-in fit

$$\check{\theta}_{\ell,\ell'} = \arg \min_{\theta} \left\| \frac{1}{|T_{\ell,\ell'}|} \sum_{i \in T_{\ell,\ell'}} g_1(W_i, \hat{\gamma}_{\ell,\ell'}, \theta) \right\|^2.\tag{159}$$

Construct $\hat{\alpha}_{\ell,\ell'}$ using the same training sample, using $\check{\theta}_{\ell,\ell'}$ when required. Evaluate the resulting baseline debiased moments on $I_{\ell'}$. For example, for $(\ell, \ell') = (1, 2)$, train on $I_3 \cup I_4 \cup I_5$ and evaluate on I_2 .

Inner baseline-only debiased estimation. Aggregate the four inner validation blocks:

$$\bar{\psi}_{1,-\ell}(\theta) := \frac{1}{n - |I_\ell|} \sum_{\substack{\ell'=1 \\ \ell' \neq \ell}}^5 \sum_{i \in I_{\ell'}} [g_1(W_i, \hat{\gamma}_{\ell, \ell'}, \theta) + \phi_1(W_i, \hat{\gamma}_{\ell, \ell'}, \hat{\alpha}_{\ell, \ell'}, \check{\theta}_{\ell, \ell'})]. \quad (160)$$

Choose a symmetric positive definite baseline weighting matrix $V_{1,n,\ell}$ using only observations in I_ℓ^c , and compute

$$\tilde{\theta}_\ell = \arg \min_{\theta} \bar{\psi}_{1,-\ell}(\theta)^T V_{1,n,\ell} \bar{\psi}_{1,-\ell}(\theta). \quad (161)$$

During this minimization, hold the inner nuisance estimates and each FSIF argument $\check{\theta}_{\ell, \ell'}$ fixed.

Outer first-step estimation. Using all observations in I_ℓ^c , construct the outer nuisance estimators $\hat{\gamma}_\ell$ and $\hat{\alpha}_\ell$, using $\tilde{\theta}_\ell$ when required. Repeat the inner and outer estimation for $\ell = 1, \dots, 5$ to obtain

$$\{(\hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell)\}_{\ell=1}^5. \quad (162)$$

Preliminary moments and optimal weighting. For each $i \in I_\ell$, evaluate

$$\hat{\psi}_i = g(W_i, \hat{\gamma}_\ell, \tilde{\theta}_\ell) + \phi(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell). \quad (163)$$

Compute

$$\tilde{\mu} = \frac{1}{n} \sum_{\ell=1}^5 \sum_{i \in I_\ell} [g_2(W_i, \hat{\gamma}_\ell, \tilde{\theta}_\ell) + \phi_2(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell)], \quad (164)$$

and construct

$$\bar{\psi} = \frac{1}{n} \sum_{i=1}^n \hat{\psi}_i, \quad \hat{\Omega}^* = \frac{1}{n} \sum_{i=1}^n (\hat{\psi}_i - \bar{\psi})(\hat{\psi}_i - \bar{\psi})^T. \quad (165)$$

Set $V_n^* = (\hat{\Omega}^*)^{-1}$.

Penalized estimation and moment selection. For each candidate coordinate $j = 1, \dots, m_2$, calculate

$$\hat{c}_j = \left([\hat{\Omega}^*]_{m_1+j, m_1+j}^2 [V_n^*]_{m_1+j, m_1+j} \right)^{1/2}. \quad (166)$$

Set

$$\hat{P}(\mu_j) = 2\hat{c}_j n^{-1} \frac{|\mu_j|}{|\tilde{\mu}_j|^2}. \quad (167)$$

Construct the augmented sample moment

$$\bar{\psi}^*(\theta, \mu) = \frac{1}{n} \sum_{\ell=1}^5 \sum_{i \in I_\ell} [g^*(W_i, \hat{\gamma}_\ell, \theta, \mu) + \phi(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell)], \quad (168)$$

and compute

$$(\hat{\theta}, \hat{\mu}) = \arg \min_{\theta, \mu} \left\{ \bar{\psi}^*(\theta, \mu)^T V_n^* \bar{\psi}^*(\theta, \mu) + \sum_{j=1}^{m_2} \hat{P}(\mu_j) \right\}. \quad (169)$$

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